# Community Networks and Entrepreneurship in a Developing Economy

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We study the role played by community-based business networks in China's transition from agriculture to domestic production in the 1990's and the subsequent transition to exporting in the 2000's. The model of occupational mobility that we develop and test incorporates a productivity enhancing role for these networks. Using administrative data covering the universe of registered firms over the 1994-2012 period and the universe of exporters over the 2002-2012 period, we provide causal evidence that networks of firms organized around the rural birth county increased the revenues of their members, both in domestic production and exporting. While this substantially increased entry into domestic production in the first transition, the incumbent domestic networks created a disincentive to enter exporting in the second transition that outweighed the positive effect of newly emerging export networks, highlighting the contrasting roles that the same networks can play at different stages of economic development.

KEYWORDS: Business networks, Entrepreneurship, Occupational mobility, Economic development.

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## 1. INTRODUCTION

The process of economic development is often characterized by an initial transition from agriculture to domestic production, followed by a second transition to higher value exporting. It is well known that community-based networks can emerge during this process to substitute for missing market institutions. Our analysis of rural-born entrepreneurs in China indicates that networks of firms from the same hometown increased the revenues of their members and thereby supported firm entry in the first transition. However, their role in the second transition turns out to be more nuanced. By successfully supporting domestic production, incumbent domestic networks raised the opportunity cost of selecting into exporting. This domestic network "overhang" more than offset the positive effect of newly emerging export networks, resulting in a net reduction in the entry of export firms in the population that we study.

The Chinese economy, which has grown at an unprecedented rate over the past decades is ideally suited to study the network dynamics described above. Its transition out of agriculture commenced in the early 1980's with the establishment of township-village enterprises (TVE's) and then accelerated with market reforms and the entry of private firms in the 1990's. A decade after market reforms commenced, China entered the WTO in 2001 and soon became the largest exporter in the world (Brandt et al., 2017). Starting with almost no private firms in 1990, there were 10 million registered private firms in 2012, accounting for 94 percent of all registered firms.[1] A striking feature of this expansion is the dominance of rural-born entrepreneurs, who managed to establish firms (usually outside their birth county) despite lacking conventional prerequisites such as higher education, inherited wealth, and access to formal credit markets.[2] The Chinese economy was largely agrarian when market reforms commenced, with approximately 2000 rural counties accounting for 74 percent of the population. Based on the administrative data that we use in this paper, 62

[1] The analysis in this paper starts in 1994, when the Chinese government reduced its commitment to support state-owned enterprises, opening the door for private firms to enter in large numbers (Zhu, 2012). It ends in 2012, when the first stage of a business registration reform commenced. The central government used firm entry as a performance indicator for local officials from that point onward and this resulted in the appearance of many shell firms in our administrative data (Barwick et al., Forthcoming).

[2] Among the rural-born entrepreneurs, 47 percent established their firm in their birth county, 11 percent in their birth prefecture but outside the birth county, 14 percent in their birth province but outside the birth prefecture, and 28 percent outside their birth province.

percent of registered firms in 2012 were run by rural-born entrepreneurs, with these firms accounting for a comparable share of registered capital.

How were the rural-born entrepreneurs able to set up their firms at an early stage of economic development when many markets were missing or incomplete? There is a common view that the Chinese government supported favored enterprises, but this would not explain the emergence of *millions* of private firms. The central thesis of our research, which is consistent with the comparative institutions literature (Peng, 2004, Allen et al., 2005, Greif and Tabellini, 2017) and the sociology literature (Honig, 1992, 1996, Goodman, 1995) is that networks of firms from the same hometown (birth county) helped their members in different ways and supported their entry into business. In particular, we expect that these networks would have allowed firms to outsource production, match with buyers and suppliers, and share information.

The productivity enhancing role of the community networks, as described above, matches the conventional characterization of inter-firm spillovers in the agglomeration literature (Combes et al., 2012, Duranton and Puga, 2020, Rosenthal and Strange, 2020). As in that literature, we specify that the spillovers are spatially concentrated: the boundary of the network is defined by the birth county-destination prefecture, there are 350 prefectures in China, and firms from a given birth county locate on average in 95 prefectures. The key difference between conventional agglomeration effects and network effects is that the former apply to all firms in a prefecture, whereas the latter are restricted to firms from a given birth county. Pre-existing social ties and accompanying social enforcement allow for higher levels of cooperation within the network. If the mutual help provided by the members of a network to each other is complementary, then firm revenue will increase with network size (net of prefecture-level agglomeration effects). The empirical analysis, described below, will verify this key relationship to establish that networks are active.

The analysis in this paper is based largely on the State Administration of Industry and Commerce (SAIC) registration database, which covers the universe of registered firms in China from the 1980's onwards. A unique feature of these administrative data is that they provide the citizenship ID of key personnel in each firm, which can be used to recover their county of birth. We designate the firm's principal (legal representative) as the "entrepreneur" for the purpose of our analysis and, based on the county of birth of each entrepreneur and the prefecture in which their firm is located, we construct approximately

160,000 birth county-destination prefecture networks from the SAIC registration database. This database can, in addition, be merged with the Customs database, which provides information on all shipments out of China to determine which firms are exporting at any point in time. While export firms may be less reliant on sub-contracting than domestic producers, owing to their need to maintain high product quality, they will still benefit from a network that provides connections to foreign buyers and export-specific information about new technologies and opportunities (Atkin and Khandelwal, 2020, Startz, 2025).[3] Our analysis thus allows for the presence of prefecture-level export networks that are restricted to export firms from the birth county. By exploiting the complete time-span of the data, which extends prior to market reforms, we can track the trajectory of each domestic and export network, from its point of inception till the endpoint of our analysis (2012). As detailed in an initial descriptive section, the prototypical network starts with a single entrant and then grows at an increasing rate over time.[4]

To interpret the preceding trajectories and to inform the empirical analysis that follows, we develop a dynamic model of occupational choice in which firms benefit from productivity-enhancing network spillovers. There are two coexisting networks in the model: a domestic network consisting of all firms from the birth county established in a particular location (prefecture) and an export network that is restricted to those firms who are engaged in that activity. The nonlinear growth trajectory that we document for both domestic networks and export networks is explained by the model as a consequence of a network multiplier (spillover) effect that grows stronger over time. The model also derives conditions under which the domestic network constrains the newly emerging export network. The key insight is that the overhang will only arise if the marginal exporter specializes in that activity and must therefore choose between domestic production and exporting. We observe such "pure" exporters in the Chinese data and their presence has also been

[3]The export firms in our analysis are all "direct exporters" who ship their product to foreign buyers and thus appear in the Customs database. These exporters are more productive than "indirect exporters," who operate through intermediaries, in China (Ahn et al., 2011). They are thus more relevant for our analysis, given its focus on the transition from domestic production to higher value exporting.

[4]The SAIC data provide the registration year and location of each firm, going back to the 1980's. The Customs database spans the entire time-span in which Chinese firms were exporting. We thus observe the precise time of commencement and the number of initial entrants for each domestic network and each export network in our data.

documented in other developing countries (Lu et al., 2014, de Astarloa et al., 2015, Blum et al., 2020).

While our ultimate objective is to quantify the contribution of the birth county networks to firm entry, with the possibility of an overhang effect, we must first establish that these networks are active. As noted above, and as implied by the model, if productivity-enhancing networks are active, then firm revenues will be increasing in network size. To implement this test, we utilize the SAIC inspection database, which provides longitudinal revenue data for a subset of registered firms from 1998 to 2012, and the Customs database, in which shipment-level data can be aggregated for each export firm in each year from 2002 to 2012. Conditional on firm fixed effects, which incorporate entrepreneurial ability, and prefecture-time period effects, which account for agglomeration effects, we estimate a positive and significant association between firm revenue and the size of the network, measured by the lagged stock of firms from the birth county that are established in the prefecture. This result is obtained for domestic producers and for exporters, with network size measured for each type of firm as described above.

While the covariates discussed above account for fixed firm characteristics and all time varying factors that affect firms in a prefecture equally, regardless of their origin, they do not control for unobserved birth county-destination prefecture productivity shocks. For example, entrepreneurs from a birth county could have preferred access to government connections in a particular prefecture that is evolving over time or could be concentrated in a sector that was growing relatively fast. The resulting increase in revenues will *pull* firms into that prefecture, with an accompanying increase in network size. The estimated network effects will be evidently biased upward. When we first-difference the estimating equation to purge the fixed effects, we will show below that the bias goes in the opposite direction. We address these potential biases by constructing statistical instruments for the *growth* in lagged network size (which becomes the endogenous variable when the estimating equation is first-differenced).

The first instrument that we construct for the growth in network size takes advantage of the fact that the birth counties are rural. Agriculture was the dominant activity in these counties as recently as the 1982 population census, with 68 percent of the workforce employed in that sector. Although this statistic declines to 37 percent in the 2000 census, agriculture continues to be a major sector. We thus construct a shift-share instrument for network size,

following Imbert et al. (2022), that is based on (negative) agricultural productivity shocks in the birth county that *push* individuals from agriculture and associated wage labor – the outside options – into non-farm business. These shocks are constructed as a weighted average of world crop-price shocks, where the weights reflect the contribution of each crop to birth county agricultural production, as well as a distance adjustment term that distributes the resulting flow of firms across destination prefectures. We draw on the recent literature, particularly Goldsmith-Pinkham et al. (2020), to implement a series of tests that verify the exogeneity of each component of the shift-share instrument.

Our second instrument exploits a unique feature of our administrative data, which is that we can pin down the precise point in time at which each birth county-destination prefecture network commenced. As noted, the prototypical network starts with a single entrant, which is consistent with previous descriptions of accidental business network formation; e.g. Damodaran (2008), Munshi (2011), Kerr and Mandorff (2023). If the initial entry shock is idiosyncratic, then the time of commencement of the network can be treated as exogenous from the perspective of the birth county as a whole. If subsequent entry is generated by a network multiplier effect, as implied by our model, then network duration will predict the growth in network size at any point in time and will be a valid instrument.

Based on our model, the additional factor that becomes relevant in the second transition is that the high profitability of domestic production, owing to the domestic network, can discourage entrepreneurs from moving into the the new export activity when it becomes available in a given prefecture. If this negative domestic network overhang is present, then any variable that is a valid instrument for the growth in domestic network size can also be used as an instrument for the growth in export network size. Domestic network duration is a particularly credible instrument in this case because the domestic network typically commences many years before the export network in a prefecture and can thus be treated as predetermined.

Our 2SLS estimates indicate that firm revenues are increasing in network size, both for domestic production and for exporting. These estimates provide causal evidence of the productivity enhancing effect of birth county networks. Since we use multiple instruments, the benchmark estimating equations are over-identified. Independent estimates of the network effects with each instrument also uncover positive and significant effects, increasing our confidence in their validity.

Having established a role for birth county networks in improving firm revenue, both in domestic production and exporting, we proceed to quantify the resulting effect of the networks on firm entry, which is the ultimate objective of the analysis, in two steps: (i) We estimate the effect of the networks on firm entry, using the model to derive the structural entry equations and to guide the choice of instruments. This exercise allows us to identify the direct positive effect of the domestic and export networks on the entry of domestic producers and exporters, respectively. It also allows us to identify the negative (overhang) effect of the domestic network on entry into exporting. (ii) We decompose the determinants of firm entry, which consist of an exogenous component, comprising the prefecture-time period effects and all unobserved factors (the error term) as well as an endogenous network component. A novel feature of this analysis is that our instrumental variable estimates allow us to quantify the contribution of the networks without placing any structure on the error term in the entry equation. We find that the domestic networks were responsible for about half of the entry of firms into domestic production. In contrast, the positive contribution of the export network to the entry of export firms is more than offset by the domestic network overhang. As a result, the entry of rural-born exporters is fairly stable over time, in contrast with the surge in the number of exporters in the Chinese economy as a whole. Atkin and Khandelwal (2020) list a number of barriers to entry into exporting in developing economies. Our analysis uncovers a new friction that has a first-order effect in this important transition on the population that we study.

Over the past 25 years, there has been an explosion of interest in the analysis of community networks (Munshi, 2014). Although there is a longstanding view in the entrepreneurship literature that networks can play an important role in supporting business formation (Birley, 1985), with a few exceptions (Kalnins and Chung, 2006, Ghani et al., 2013, Kerr and Mandorff, 2023), this literature does not attempt to estimate network effects directly. In contrast, many studies have attempted to estimate the effect of migrant networks on the labor market outcomes of their members. Occupational mobility, which is the subject of our analysis, and spatial mobility are closely related, both conceptually and in practice. For example, with the exception of the birth prefecture network, all other networks in our analysis require entrepreneurs to migrate. It is consequently not surprising that the statistical instruments deployed in the migration literature are broadly related to the instruments that we use: (i) shocks at the origin that push individuals to migrate (Munshi, 2003, Karadja

and Prawitz, 2019, Imbert et al., 2022, Tang, 2024), and (ii) historical conditions that determine current network size (Woodruff and Zenteno, 2007, McKenzie and Rapoport, 2007, 2010, Patel and Vella, 2013, Spitzer and Zimran, 2023). With the recent availability of large administrative data, extensive mobile phone data, and rich (digitized) historical data, there continues to be interest in the analysis of migrant community networks. However, recent studies that use these data; e.g. Stuart and Taylor (2021), Barwick et al. (2023), Blumenstock et al. (2025) have attempted to identify network effects without statistical instruments. Our analysis, which uses a model-based approach to understand the biases that arise when estimating network effects and to construct statistical instruments in the event that the orthogonality condition is violated, serves as a guide for future studies using large-scale data.

The findings in this paper also speak to two influential literatures in macro development: First, a large body of research, starting with Acemoglu et al. (2001) points to the long shadow of historical (pre-modern) institutions on the functioning of contemporary legal and political systems, which are seen to be essential for economic growth. As some critics have noted, this view does not adequately explain the remarkable Chinese growth experience since the 1990's, which occurred without secure property rights, rule of law, and protection against government power. Our analysis fills in the missing piece by showing that informal networks that emerge at early stages of economic development can facilitate firm formation and growth in the absence of these formal institutions. Second, a voluminous literature, going back to Galor and Zeira (1993) and Banerjee and Newman (1993), studies how market imperfections constrain occupational mobility in developing economies, resulting in the persistence of inequality. Our analysis indicates that community-based networks can break these occupational traps. At the same time, our unique administrative data, coupled with the compressed nature of the Chinese development experience, reveals that the same domestic networks that support occupational mobility at an initial stage of the development process, end up restricting further upward mobility at a later stage.

The paper is organized as follows: Section 2 describes the birth county-destination prefecture networks and their trajectories. Section 3 develops a dynamic model of occupational choice that adds a network component and a trade component to the Roy (1951) model. The tension between the domestic network and the export network is an important feature of this model. Section 4 maps the model to the data, deriving the firm revenue and firm entry

equations that we estimate. Section 5 documents the causal effect of network size on firm revenue, separately in domestic production and exporting. Section 6 quantifies the effect of the networks on firm entry and Section 7 concludes with a discussion on efficiency and the welfare consequences of interventionist policies.

## 2. BIRTH COUNTY NETWORKS

### 2.1. *Network Domain*

Ethnicity in China is defined by the rural home town or birth county (Honig, 1992, 1996, Goodman, 1995) and it is well documented that *laoxiang* or "native-place fellows" help each other in different ways (Ma and Xiang, 1998, Zhang and Xie, 2013).[5] Chambers of commerce that bring entrepreneurs from the same origin together (*yidi shanghui*) are also commonly found in Chinese cities. As with agglomeration effects, the help provided by networks of firms, such as connections and information, is inherently local. We thus specify that the network consists of firms from the same birth county operating in a given destination prefecture. Each prefecture consists of an urban center and eight counties on average, and there are approximately 350 prefectures in China. Many government infrastructure and investment initiatives are organized at this administrative level, and buyer and sellers tend to locate in prefecture-level cities, so the birth county-destination prefecture would appear to be the appropriate spatial domain for the networks that we study.

Although our administrative data do not report how production is organized, we assume that domestic producers benefit from the information, connections, and other forms of support that they receive from all other firms. While exporters may be more vertically integrated and less reliant on out-sourcing than domestic producers, they will also benefit from an export-specific network, as noted in the previous section. The domestic network is thus specified to include all firms from the birth county who are located in the prefecture, whereas the export network is restricted to exporting firms. We will provide empirical support for this specification of the network domains below, documenting that firm revenues are determined by the size of the relevant network, and that cross-network effects are absent.

[5] In Chinese cities, migrant enclaves are often named after a sending province, but as Ma and Xiang (1998) note, this nomenclature is misleading because the enclave typically consists of individuals from a single rural county or two neighboring counties. In this paper, we use the terms hometown and birth county interchangeably.

The preceding definition of the networks distinguishes between activities – domestic production versus exporting – but not between sectors within these activities. As seen in Appendix Table A1, Column 1, 67 percent of the firms in a given birth county-destination prefecture-time period operate in a single most popular 3-digit Input-Output industry. An additional 26 percent are set up in related upstream-downstream and complementary industries.[6] The preceding statistics are derived from the SAIC registration database, and we can construct corresponding statistics for the export network by combining the SAIC data with the Customs database. As seen in Appendix Table A1, Column 2, the statistics for the export network are 80 percent and 17 percent. It follows that most firms from a birth county in a given prefecture at a given point in time operate in the same or related industries. We allow for spillovers across these industries, and for common factors such as access to credit, by ignoring sectors when constructing the networks in our analysis.

### 2.2. *Network Trajectories*

We assume that a given birth county-destination prefecture network has commenced if we observe at least one firm from the birth county in that prefecture for three consecutive years.[7] The SAIC data, which extend back to the 1980's, provide the registration year and the location of each firm. We can thus observe the precise point in time at which each network commenced. As seen in Appendix Table A2, there are 166,117 domestic networks and 11,365 export networks in our data. Some of these networks would have started with a single firm and stayed that way. While the term "network" may be somewhat of a misnomer in this case, we retain these singleton networks in the estimation sample to avoid selection bias. Our primary interest, however, is in the multi-firm networks and, as can be seen in the table, 99 percent of domestic firms and 93 percent of export firms belong to such networks.

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[6] We use the 2007 input-output table from the Chinese National Bureau of Statistics to determine whether any two industries are upstream-downstream or complementary. An industry is defined as being upstream or downstream of another industry if its input or output share (derived from the input-output table) exceeds 0.01. Two industries are defined as being complements if the average correlation coefficient of their input-output shares, across upstream-downstream industries, exceeds 0.2. This methodology is based on Fan and Lang (2000).

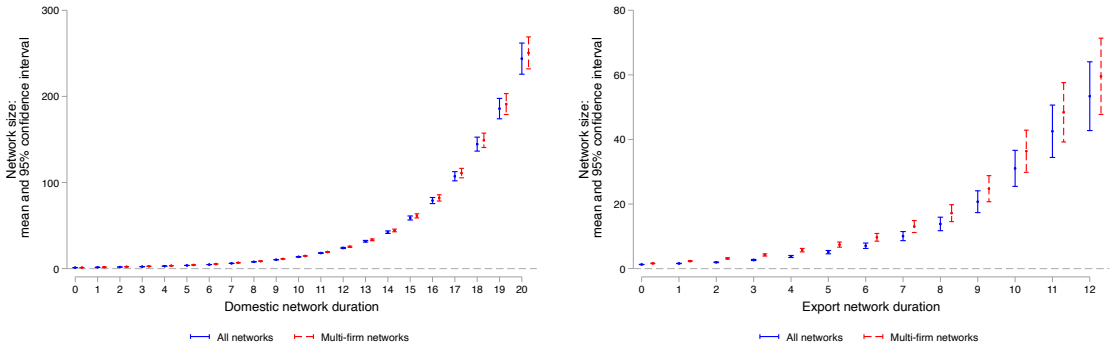
[7] When we test for network effects below, we will include the lagged stock of firms from the birth county who are active in the prefecture in the estimating equation. We will first-difference that equation to purge fixed effects, which adds a second lag. To include a birth county-destination prefecture when estimating network effects, at least one firm must therefore be present for three consecutive years.

Moreover, 85 percent of the multi-firm domestic networks and 76 percent of the multi-firm export networks commenced with a single firm.

Our interpretation of the preceding statistics is that most firms belong to multi-firm networks that start with a single firm. This pioneering entrepreneur presumably receives a fortuitous opportunity in a particular location (prefecture). Once the pioneer's firm is set up, other entrepreneurs from the birth county follow and the network grows in size over time. Figure 1 provides graphical support for the preceding characterization of the network's evolution by plotting network size against network duration (time after commencement). The vertical lines in the figure measure the dispersion in network size (the 95 percent confidence interval) and their midpoints denote the mean. As can be seen, the mean and the dispersion in network size are increasing at an increasing rate with duration, both for domestic and for export networks. The average number of initial entrants is 1.15 for domestic networks and 1.3 for export networks. The dispersion in the number of initial entrants is extremely small, despite the fact that networks commence at very different points in time, which is consistent with our characterization of network formation. The subsequent nonlinear increase in (mean) network size is consistent with a network multiplier effect, which is a key feature of the model that follows. Notice that the trajectories for all networks and for multi-firm networks match closely in the figure. We will use network duration to predict the growth in network size when estimating network effects. While we estimate this relationship for all networks, it is driven by the multi-firm networks.

There were approximately 4.5 million registered firms and 59 thousand exporting firms, drawn from rural birth counties, in 2012.[8] The enormous difference in the total number of firms does not imply, however, that there is no tension between the domestic and export networks. As seen in Figure 1, average domestic network size is 244, 20 years after commencement. The corresponding statistic for export networks, 12 years after commencement, is 53. In those birth county-destination prefectures where both types of networks co-exist, export firms make up a non-negligible share of all firms.

[8] Our analysis, drawing on the Customs database, is based on exporters who ship their output directly to foreign buyers. Rural-born direct exporters account for 30 percent of all direct exporters and 50 percent of domestically owned direct exporters.




(a) Domestic Network (b) Export Network

Source: SAIC registration data and Customs data.
Duration is measured by the time (in years) after the network's commencement.
Network size is measured by the number of firms from the birth county in the destination prefecture.
The domestic network includes all firms, while the export network is restricted to exporting firms.

FIGURE 1.—The Trajectory of the Network

## 2.3. *Birth County Ties*

As documented above, most entrepreneurs belong to multi-firm networks; i.e. there are multiple firms from the birth county in the prefectures where they are located. Restricting attention to such entrepreneurs, who were (in addition) active in 2012, we find that 66 percent of the domestic producers and 46 percent of the exporters were located outside their birth county. As many of these entrepreneurs would have moved a long time ago, a natural question to ask is whether they remain connected to their natal community in some important way.

Table I provides descriptive support for the importance of birth county ties for entrepreneurs who are (i) located outside their natal county, (ii) belong to multi-firm networks, and (iii) were active in 2012, the end point of our analysis. In addition to the entrepreneur (legal representative), the SAIC registration database also lists other key personnel in the firm.[9] Columns 1-2 report the fraction of these individuals who are born in the same county as the legal representative, as well as the counterfactual fraction that is constructed by randomly assigning listed individuals (other than the legal representative) across firms in

[9]The legal representative, who has the authority to enter into binding obligations on behalf of the company, typically functions as the firm's president, chairman or proprietor. Other listed individuals include directors, senior managers, and external "supervisors." Firms with a single listed individual, who will necessarily be the legal representative, are excluded from the analysis in this section.

the prefecture. Focusing on the domestic producers in Column 1, we see that half of the listed individuals are born in the same county as the legal representative, which is 50 times more than what would be obtained by random assignment. Qualitatively similar results are obtained in Column 2 with export firms, except that the gap between the observed and counter-factual statistics is not quite as large.

Table I, Columns 3-4 assess the strength of the birth county ties in a different way, by examining the links between firms in the prefecture. While networks largely rely on informal interactions between socially connected firms, formal links can also be used to complement these interactions and increase cooperation. We denote two firms (located in the same prefecture) as being "linked" if the same individual is listed in both of them. Based on this definition, we see that 50 percent of linked domestic producers are linked to firms from the same birth county. Once again, this is approximately 50 times more than what would be obtained if firms with links were randomly matched in the prefecture. Column 4 reports the corresponding statistics for export firms with qualitatively similar results.

TABLE I

BIRTH COUNTY TIES

| Variable: | fraction of key personnel from the entrepreneur's birth county | | fraction of linked firms that are linked to a firm from the same birth county | |
|---|---|---|---|---|
| Type of firm: | domestic producer | exporter | domestic producer | exporter |
| | (1) | (2) | (3) | (4) |
| Observed mean | 0.515 | 0.535 | 0.515 | 0.691 |
| Counter-factual mean | 0.009 | 0.023 | 0.011 | 0.100 |

Note: statistics are computed using SAIC registration data and Customs data.
The sample is restricted to firms established outside their birth counties that belong to multi-firm networks and were active in 2012.
The legal representative is denoted as the "entrepreneur" in our analysis.
Linked firms have at least one key person in common.
The counter-factual mean is based on the random assignment of key personnel and the random matching of linked firms in the prefectures where they are located.

The evidence presented in Table I indicates that entrepreneurs remain connected to their birth county, even when their firms are established elsewhere. Moreover, their firms are disproportionately linked to firms from the same birth county in the prefectures where they are located. We expect the birth county homophily we have uncovered to extend to informal

interactions between firms, which, in turn, will further support mutual help and cooperation. The model of entrepreneurship (occupational choice) that follows will thus incorporate a productivity enhancing role for birth county networks.

## 3. THE MODEL

Since networks are organized at the level of the birth county-destination prefecture, and the dynamics we are interested in occur at this level, the analytical model that we present in this section is based on a single (rural) birth county and a single destination prefecture where businesses are established. The model describes the dynamics of occupational choice and firm entry when networks are active. It also derives conditions under which the domestic network overhang dampens entry into exporting.

### 3.1. *Population and Technology*

Successive cohorts of agents, indexed by $t' = 1, \ldots, T$ are born in the county. All agents continue to live until the terminal date $T$. The aggregate measure of agents in each cohort is $s$ and the ability $\omega$ of each agent in the cohort is drawn from an i.i.d. log uniform distribution: $\log \omega$ is uniformly distributed on $[A-1, A]$. Cohort $t'$ agents who enter the workforce in period $t'$ choose occupations at each date $t \geq t'$.

There are two possible occupations: a traditional occupation and entrepreneurship. An agent of ability $\omega$ earns a stationary payoff $\omega^\sigma$ in the traditional occupation at each date, where $\sigma \in (0, 1)$. If he chooses to become an entrepreneur, he can produce either for the domestic ($d$) market or the export ($e$) market, or both. Serving a market $b \in \{d, e\}$ requires investing in a plant specific to that market, with capital size $K_{bt}$ at date $t$. Investments in either type of plant are irreversible: capital already invested cannot be disinvested, while it is possible to invest more at later dates. Hence, an entrepreneur is committed to a market $b$ once he invests in it. The capital irreversibility constraint is $K_{bt} \geq K_{b,t-1}$ for all $t$.[10]

[10]The irreversibility assumption is reasonable in the context of a rapidly growing economy at early stages of economic development. Although there is no exit in our model, all of the results that follow would be retained if we allowed for a uniform and exogenous death rate. The empirical analysis in this paper is thus based on the stock of *surviving* firms.

A plant of size $K_{bt}$ owned by an entrepreneur of ability $\omega$ generates revenues at $t$:

$$R_{dt} = C_{dt}\omega^{1-\alpha}K_{dt}^{\alpha}, R_{et} = C_{et}\omega^{\delta(1-\alpha)}K_{et}^{\alpha}, \tag{1}$$

where $\alpha \in (0,1)$ reflects diminishing returns to size and $\delta > 1$ represents an ability premium on the export market. TFP (or revenue productivity, to be more precise) depends on the entrepreneur's individual ability $\omega$ and a productivity multiplier $C_{bt}$, which is comprised of an endogenously determined birth county-destination prefecture network effect:

$$C_{dt} = [n_{t-1}]^{\theta_d}, C_{et} = [n_{e,t-1}]^{\theta_e} \tag{2}$$

where $n_{t-1}$ measures the stock of firms originating from the birth county and located in the prefecture in period $t-1$, and $n_{e,t-1}$ measures the corresponding stock of export firms. The assumption in this formulation of the network effects is that help at any date is provided to network members by experienced incumbents who have been operating for at least one period already. As discussed in the preceding section, the domestic network includes all firms, whereas the export network is restricted to export firms. Parameters $\theta_d, \theta_e$ denote elasticities of TFP with respect to network size for domestic producers and exporters, respectively. Note that the nonlinear specification of the network function accomodates congestion effects, which have previously been documented in labor market networks (Beaman, 2012, Tang, 2024).

Capital costs for domestic and export plants are as follows:

$$E_{dt} = rK_{dt}, E_{et} = r(1+I)K_{et} \tag{3}$$

where $r$ includes interest and material costs of equipment, and $I > 0$ is the incremental cost of operating an export plant, arising from the need to vertically integrate production or to conform to international quality standards.[11] An important additional feature of the model is the presence of diseconomies of scope, incurred by *mixed exporters* who produce for both the domestic and the export market. This diseconomy of scope, which could be

[11] We could add a labor input to the production function, without changing the results that follow, as long as all firms face a common wage.

reformulated as a managerial technology with diminishing returns to "span of control," as in Lucas (1978), is represented by a fixed cost $\beta$ in addition to plant costs (3). Hence, the total cost of a mixed exporter equals $E_{dt} + E_{et} + \beta$. This will result in the presence of *pure exporters*, who specialize in that activity.[12]

Given the irreversibility of market entry decisions, network sizes cannot shrink: $n_t \geq n_{t-1}$, $n_{et} \geq n_{e,t-1}$. With regard to exogenous initial conditions, we make the simplifying assumption that there are $n_0$ entrants at date $0$, $n_{e0}$ of whom are exporters. Later we shall discuss the consequences of export opportunities starting at some later date $\hat{t} > 1$, so that agents only have the option to enter the domestic market between dates 1 and $\hat{t} - 1$.

### 3.2. *Occupational Choice in Equilibrium*

To simplify the exposition, we assume that agents are myopic and that network sizes at past dates are observable by all agents. As shown in Appendix B.2, the results that follow extend to the case where agents are forward looking but discount future profits at a high enough rate. Consider date $t$, with given productivity multiplier $C_{bt}, b \in \{d, e\}$. An entrepreneur of ability $\omega$ who was active in previous periods inherits plant sizes $K_{b,t-1}$ and selects current plant sizes $K_{bt}, b \in \{d, e\}$ to maximize

$$[C_{dt}\omega^{1-\alpha}K_{dt}^{\alpha} - rK_{dt}] + [C_{et}\omega^{\delta(1-\alpha)}K_{et}^{\alpha} - r(1+I)K_{et}] - \beta\mathbb{I}(K_{dt}K_{et}) \tag{4}$$

subject to the irreversibility constraints

$$K_{bt} \geq K_{b,t-1}, b \in \{d, e\} \tag{5}$$

where $C_{bt}$ is given by (2), $\mathbb{I}(x)$ denotes an indicator function which takes the value $1$ if $x > 0$ and $0$ otherwise, and past plant size is set equal to zero for any entrepreneur that has not entered the corresponding market previously.

Recall that network sizes are non-decreasing. This implies that the productivity multiplier $C_{bt}, b \in \{d, e\}$, is increasing over time and, hence, that optimal plant sizes must increase over time for incumbents. It follows that the irreversibility constraint is non-binding

[12] Lu et al. (2014) also use differences in fixed costs to motivate the coexistence of domestic producers, pure exporters, and mixed exporters.

with respect to plant size choice, within any market. Maximizing profit, as expressed in equation (4), with respect to current plant size in each market and then substituting back in the profit function, the equilibrium profit (conditional on entry) for an entrepreneur with ability $\omega$ in period $t$ can then be derived for each occupation $W \in \{O, D, E, M\}$, where $O$ refers to the traditional (other) occupation, $D$ is domestic production, $E$ is pure exporting, and $M$ is mixed exporting:

$$\Pi_{Ot}(\omega) = \omega^{\sigma} \tag{6}$$

$$\Pi_{Dt}(\omega) = \omega[\frac{1}{\zeta}]C_{dt}^{\frac{1}{1-\alpha}} \tag{7}$$

$$\Pi_{Et}(\omega) = \omega^{\delta}[\frac{1}{\zeta\gamma}]C_{et}^{\frac{1}{1-\alpha}} \tag{8}$$

$$\Pi_{Mt}(\omega) = \Pi_{Dt}(\omega) + \Pi_{Et}(\omega) - \beta \tag{9}$$

where $\zeta \equiv \frac{r^{\frac{\alpha}{1-\alpha}}}{\alpha^{\frac{\alpha}{1-\alpha}} - \alpha^{\frac{1}{1-\alpha}}}$ and $\gamma \equiv (1+I)^{\frac{\alpha}{1-\alpha}}$.

The above profits are generated by optimal choices on the intensive margin, for a given occupational choice on the extensive margin. We now turn to equilibrium (extensive form) occupational choices. Observe from (6) - (9) that there are three ability thresholds: $\omega_{dt}^*$ at which individuals are indifferent between the traditional occupation and domestic production, $\omega_{et}^*$ at which individuals are indifferent between domestic production and exporting, and $\omega_{mt}^*$ at which individuals are indifferent between pure exporting and mixed exporting.

$$\log \omega_{dt}^* = \frac{\log \zeta}{1-\sigma} - \frac{\log C_{dt}}{(1-\alpha)(1-\sigma)} \tag{10}$$

$$\log \omega_{et}^* = \frac{1}{\delta - 1}\left[\log \gamma + \frac{\log C_{dt} - \log C_{et}}{1-\alpha}\right] \tag{11}$$

$$\log \omega_{mt}^* = \log \beta + \log \zeta - \frac{\log C_{dt}}{1-\alpha} \tag{12}$$

The lower threshold $\omega_{dt}^*$ is decreasing in the size of the domestic network ($C_{dt}$) and increasing in the cost of capital ($\zeta$) and the returns to ability in the traditional occupation ($\sigma$). The middle threshold $\omega_{et}^*$ is decreasing in the size of the export network ($C_{et}$) and the

ability premium in exporting ($\delta$) and increasing in the size of the domestic network ($C_{dt}$) and the incremental cost of exporting ($\gamma$). The higher threshold $\omega^*_{mt}$ is decreasing in the size of the domestic network ($C_{dt}$) and increasing in the scope diseconomy parameter ($\beta$) and the cost of capital ($\zeta$).

As shown in Appendix B.3, firms of all three types $D, E, M$ co-exist in China. We derive conditions on the model's parameters that ensure that the ability thresholds are appropriately ordered and, hence, that this is true at the first date that any network is active and at any subsequent date.

PROPOSITION 1: *Parametric conditions derived in Appendix B.1 ensure that there is a unique equilibrium in which a positive mass of agents from cohort $t'$ sort into each of the four occupations at any date $t \geq t'$. This equilibrium is characterized by ability thresholds*

$$A - 1 < \log \omega^*_{dt} < \log \omega^*_{et'} < \log \omega^*_{mt} < A \tag{13}$$

*such that*
*(a) those with ability below $\omega^*_{dt}$ stay in the traditional occupation ($O$)*
*(b) those between $\omega^*_{dt}$ and $\omega^*_{et'}$ specialize in domestic production ($D$)*
*(c) those between $\omega^*_{et'}$ and $\omega^*_{mt}$ specialize in exports ($E$)*
*(d) those above $\omega^*_{mt}$ serve both markets ($M$).*

The proof of the proposition is in Appendix B.1. Note that the threshold $\omega^*_{et'}$ for transition between domestic production and pure exporting for cohort $t'$ remains unchanged at every subsequent date $t > t'$. This owes to the irreversibility of past entry decisions: agents from cohort $t'$ who choose to export at $t'$ the very first date they come into being, are locked into exporting at all subsequent dates. So they cannot switch to specializing in domestic production at later dates if that becomes more profitable than export specialization. Similarly, those firms that selected into domestic production at $t'$ cannot become pure exporters at a later date.

The domestic production threshold $\omega^*_{dt}$ and the mixed export threshold $\omega^*_{mt}$ are not similarly constrained by past decisions. We will see below that domestic network size $n_t$ expands over time. As a result, some agents from cohort $t'$ who started in the traditional occupation will move into domestic production at older ages, and some pure exporters will

become mixed exporters (by adding a domestic plant once it becomes sufficiently remunerative). While this implies that $\omega^*_{dt}$, $\omega^*_{mt}$ are falling in $t$, the parametric conditions in Appendix B.1 ensure that a positive mass of agents from cohort $t'$ remain in the traditional occupation and in pure exporting at all dates.

### 3.3. *Firm Entry*

From Proposition 1, individuals with ability $\log\omega \in [\log\omega^*_{dt}, A]$ become entrepreneurs. Based on the expression for $\omega^*_{dt}$ from (10) and unpacking $C_{dt}$:

$$n_t = ts[A - \log\omega^*_{dt}] = ts[A - \frac{\log\zeta}{1-\sigma} + \frac{\theta_d \log n_{t-1}}{(1-\sigma)(1-\alpha)}] \tag{14}$$

Recall that $t \cdot s$ is the stock of potential entrepreneurs from the birth county in period $t$. Starting with $n_0$ entrants in period 0, we can solve equation (14) recursively, from one period to the next, to derive the stock of firms $n_t$ in any period $t$. $n_t$ will evidently be increasing (nonlinearly) in $t$ and $n_0$, on account of the network multiplier effect. This is consistent with the trajectory of the representative domestic network with respect to its duration, as reported in Figure 1a.

Based on Proposition 1, individuals from cohort $t'$ with ability $\log\omega \in [\log\omega^*_{et'}, A]$ become exporters. There is no further entry into exporting from the $t'$ cohort after that period.[13] Thus, the stock of exporters at any period $t$ is just the sum of exporters supplied by all preceding cohorts. The marginal pure exporter in cohort $t'$, with ability $\omega^*_{et'}$, is indifferent between domestic production and pure exporting. Following the same steps as above, we derive $\omega^*_{et'}$ from (11) and then unpack $C_{dt'}$, $C_{et'}$ to obtain:

$$n_{et} = ts[A - \frac{\log\gamma}{\delta - 1}] + \frac{s}{(\delta-1)(1-\alpha)} \sum_{t'=1}^{t} [\theta_e \log n_{e,t'-1} - \theta_d \log n_{t'-1}] \tag{15}$$

There is a (export) network multiplier in this equation as well, which explains the highly nonlinear export network trajectory in Figure 1b. However, an an additional factor must

[13]The transition from domestic production to exporting does not arise in the analytical model where both networks have been assumed to start at the same time. In the next section, we assume that export opportunities arrive after the domestic network has started, in which case some high ability domestic producers who did not have access to the export network when they entered will become mixed exporters, as in Melitz (2003).

be considered with export entry, which is that the ability threshold $\omega^*_{et'}$ that pins down the number of exporters supplied by cohort $t'$ is determined by network sizes $(n_{e,t'-1}, n_{t'-1})$, in exporting as well as domestic production. In particular, an increase in domestic network size, $n_{t'-1}$, increases $\omega^*_{et'}$ and, thus, results in a decline in the number of exporters.

The (domestic) network overhang, which we identify and quantify in the sections that follow, arises because the marginal exporter is a pure exporter who must choose between domestic production and exporting. Pure exporters have been observed in many developing countries and we document their presence in China by matching the economic censuses to the Customs database in Appendix B.3. Pure exporters comprise around 15 percent of all exporters and their revenues lie between domestic-firm revenues and mixed-exporter revenues, as implied by the model. Although the number of pure exporters may not be substantial, they are critical to the analysis. If the marginal exporter were a mixed exporter, instead, then the domestic network would not negatively impact entry into exporting.

The root cause of the network overhang in our model is the scope diseconomy, which introduces a nonseparability between domestic production and exporting. As a result, most active entrepreneurs, with the exception of the mixed exporters, must choose between these activities. The nonseparability does not arise in the Melitz (2003) model, in which the entire overhead production cost is accounted for in domestic profits and, hence, a firm will export if its additional revenues exceed the additional costs. This implies that a demand shock on the domestic market will have no bearing on the firm's export decision. However, the nonseparability does arise in Fan et al. (2020) and Almunia et al. (2021), who extend the Melitz model by allowing for increasing marginal costs. Positive shocks on the domestic market now reduce the firm's exports. While the focus of these recent papers is on the intensive margin, they could, in principle, generate the same tradeoff at the extensive margin between domestic production and exporting as in our model.

## 4. ESTIMATING EQUATIONS

We will begin the empirical analysis by testing the mechanism underlying the model, which is that a firm's performance, measured by its revenue, is increasing in the number of firms from the birth county that are established in the same prefecture. We will then proceed to quantify the effect of the birth county networks on firm entry. In this section, we derive the revenue estimating equations and the entry estimating equations from the model.

To map the analytical model to the data, we will need to augment it in a number of ways. Some of these extensions will generate bias in the equations that we estimate, while others will motivate the construction of instruments that address these biases.

### 4.1. *Mapping the Model to the Data*

**1.** In the model, cohorts of fixed size, $s$, enter in each period and start making occupational choices immediately. There is no exit (until the terminal date $T$) and, hence, the number of active workers in period $t$ is $s \cdot t$. When we take the model to the data, we assume, instead, that individuals enter the workforce at age $\underline{a}$ and exit at $\overline{a} - 1$. While the number of active entrepreneurs (firms) may be increasing rapidly over time at early stages of economic development, the size of the workforce, $s_t$, consisting of all individuals in the $[\underline{a}, \overline{a} - 1]$ age range in period $t$, evolves relatively slowly. We thus assume that all $a \equiv \overline{a} - \underline{a}$ age cohorts are (approximately) equally sized at any point in time to simplify the analysis that follows.

**2.** We allow for multiple birth counties and multiple destination prefectures, indexed by $j$ and $k$, respectively. We do this in a very simple way, by assuming that $1/K$ agents from any birth county $j$ have the opportunity to become entrepreneurs at a given destination prefecture $k$ (with $K$ denoting the total number of available destinations). A network will only form in destination $k$ if an individual from $j$ receives a fortuitous opportunity there. If that event does occur, then the dynamics of the model apply to that particular birth county-destination prefecture pair.

**3.** For a given birth county-destination prefecture pair, we allow the domestic network to start exogenously at time $t_{djk}$ and for the export network to start exogenously at a later date $t_{ejk} \geq t_{djk}$.

**4.** We add a time varying TFP component, $U_{jt}$, to the payoff in the traditional occupation in the birth county, which is now specified as $U_{jt}\omega^{\sigma}$. Denote $u_{jt} \equiv \log U_{jt}$.

**5.** We augment the productivity multiplier $C_{bjkt}$, $b \in \{d, e\}$, by including a market-time effect $Q_{bkt}$, which incorporates agglomeration effects and other factors such as product demand and government support that apply equally to all firms in prefecture $k$, as well as a productivity shock $V_{bjkt}$ that is specific to firms from birth county

$j$: $C_{djkt} = V_{djkt}Q_{dkt}[n_{jk,t-1}]^{\theta_d}$, $C_{ejkt} = V_{ejkt}Q_{ekt}[n_{ejk,t-1}]^{\theta_e}$. Denote $q_{bkt} \equiv \log Q_{bkt}$, $v_{bjkt} \equiv \log V_{bjkt}$.

### 4.2. *Domestic production*

Based on the model, the revenue obtained by a domestic producer with ability $\omega$, $R_{djkt} = C_{djkt}\omega^{1-\alpha}K_{djkt}^{\alpha}$. Taking logs, substituting the value of the profit maximizing capital investment, and unpacking $C_{djkt}$:

$$\log R_{djkt} = \frac{\alpha}{1-\alpha}\log\left(\frac{\alpha}{r}\right) + \frac{q_{dkt}}{1-\alpha} + \frac{\theta_d \log n_{jk,t-1}}{1-\alpha} + \log\omega + \frac{v_{djkt}}{1-\alpha} \tag{16}$$

Leaving aside the constant, the equation derived above consists of four terms: a market-time term, a network term, an ability term, and an unobserved productivity term. If firm-level information is available over time, then firm fixed effects can be included in the estimating equation to collect the unobserved ability term. If firms from a given birth county are established in multiple destination prefectures and firms from many origin counties are established in a given prefecture, then it is possible to empirically disentangle network effects from market-time effects, with the latter being subsumed in destination-time period effects that we also include in the estimating equation. Conditional on the firm fixed effects and the destination-time period effects, consistent estimates of the network effects will be obtained if network size, $n_{jk,t-1}$, is uncorrelated with the error term $v_{djkt}$. The entry equation that we derive next will allow us to systematically examine this orthogonality condition.

We define the entrepreneurial propensity by the number of firms divided by the number of potential entrepreneurs: $\frac{n_t}{ts}$ in equation (14). Modifying that equation, based on the discussion in the preceding section:

$$\frac{n_{jkt}}{s_{jt}} = A_{jk} + \frac{q_{dkt}}{(1-\sigma)(1-\alpha)} + \theta_d\frac{\log n_{jk,t-1}}{(1-\sigma)(1-\alpha)} + \frac{v_{djkt}}{(1-\sigma)(1-\alpha)} - \frac{u_{jkt}}{1-\sigma} \tag{17}$$

By moving the number of potential entrepreneurs, $s_{jt}$, to the denominator of the left hand side of the estimating equation, we are left with a specification that has a similar structure

to the revenue equation derived above.[14] There are five terms in this equation: a fixed birth county-destination prefecture term, which more flexibly incorporates the ability parameter $A$ in the model, a market-time term, a network term, and two productivity terms. Notice that the TFP term, $u_{jt}$, is now specified at the birth county-destination prefecture-time period level, as $u_{jkt}$. As observed in equation (17), a decline in TFP in the traditional occupation, which is the outside option to business, has a positive effect on the stock of firms. This effect is exogenously distributed across destinations, in the same way that a fixed fraction of agents from the birth county are assumed to have the opportunity to become entrepreneurs in a given destination prefecture.

Notice that the productivity shock, $v_{djkt}$, appears in the error term of equations (16) and (17). When we estimate these equations, we will first-difference them to purge fixed effects. The endogenous variable then becomes the (lagged) growth in network size, $\log n_{jk,t-1} - \log n_{jk,t-2}$, and the error term now contains $v_{djkt} - v_{djk,t-1}$. It is evident that the $v_{djk,t-1}$ term (with a minus sign) in the residual of the first-differenced equations will bias the estimated network effects downwards.[15] In addition, when we first-difference equation (17), $n_{jk,t-1}$ appears on both sides of the estimating equation, biasing the network effect even further downward if network sizes are measured with error.

An additional source of bias is present if $v_{djkt}$ is (positively) serially correlated. This is true even if equations (16) and (17) are estimated in levels, without first-differencing, because $n_{jk,t-1}$ is positively correlated with $v_{djk,t-1}$ from the latter equation and, hence, $\log n_{jk,t-1}$ is positively correlated with $v_{djkt}$. This positive bias works in opposition to the negative bias described above and, hence, the sign of the net bias is ambiguous. Regardless of the sign, it is necessary to construct statistical instruments for the growth in network size when estimating equations (16) and (17).

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[14]The denominator on the left hand side of equation (17) should, more correctly, be $s_{jt}/K$. This could be accomodated by multiplying the right hand side by $1/K$, but this has no bearing on our interpretation of the estimates that follow in Section 5 or the quantitative analysis in Section 6. We similarly ignore the $1/K$ multiplier when we specify the export propensity equations below.

[15]Lagging (17) by one period, we see that $n_{jk,t-1}$ and $v_{djk,t-1}$ are positively correlated, hence the negative bias since $v_{djkt} - v_{djk,t-1}$ appears in the error term of the first-differenced equations. This negative bias is conceptually related to the well known Nickell bias, which arises when estimating dynamic panel models with fixed effects.

The first instrument that we propose for the growth in domestic network size is based on the observation that TFP shocks to the payoff in the traditional (agricultural) occupation in the birth county, $u_{jkt} - u_{jk,t-1}$, will determine firm entry in each destination prefecture, $n_{jkt} - n_{jk,t-1}$, as seen in equation (17). These shocks will push potential entrepreneurs into business and are thus uncorrelated with the unobserved (non-farm) productivity shocks in the destination prefecture, $v_{djkt} - v_{djk,t-1}$, that pull them into that activity. Although the instrument based on agricultural shocks is plausibly exogenous, one limitation is that it could potentially apply to multiple coexisting networks. For example, a negative agricultural productivity shock in the birth county could push individuals into non-farm wage labor and business, making it difficult to disentangle (overlapping) labor market and business network effects. This limitation does not, however, apply to the instrument that we propose next.

Our second instrument is motivated by previous descriptions of business network formation and by the network trajectories documented in Section 2, which indicate that there is a fortuitous, idiosyncratic, aspect to initial entry. The timing of initial entry and, by extension, the duration of the network at any subsequent point in time, can then be treated as exogenously determined. The network's duration will, in addition, determine its contemporaneous growth rate, on account of the multiplier effect described in Section 3, thus satisfying the conditions for a valid statistical instrument. We expect that this novel instrument will be especially useful when estimating network effects with digitized administrative and historical data that are now becoming available because these data could potentially be extensive enough to pin down the precise point in time when each network commences.

Two instruments are available, as discussed above, when we estimate equation (16) to identify domestic network effects. To complete the analysis, however, we will also want to quantify the effect of the domestic network on firm entry by estimating equation (17). Notice that only the second – domestic network duration – instrument will be valid when we estimate that equation because $u_{jkt}$ appears in the error term.

### 4.3. *Exporting*

The expression for export revenue, corresponding to the domestic revenue equation (16), is derived from the model as:

$$\log R_{ejkt} = \frac{\alpha}{1-\alpha}\log\left(\frac{\alpha}{r(1+I)}\right) + \frac{q_{ekt}}{1-\alpha} + \frac{\theta_e \log n_{ejk,t-1}}{1-\alpha} + \delta\log\omega + \frac{v_{ejkt}}{1-\alpha} \quad (18)$$

If firm-level export revenues are available over time, then firm fixed effects can be included in the estimating equation to account for the ability term. As with equation (16), destination-time period effects will account for the market-time term. Once we first-difference equation (18) to purge the fixed effects, consistent estimates of the network effects will be obtained if the growth in network size, $\log n_{ejk,t-1} - \log n_{ejk,t-2}$, is uncorrelated with $v_{ejkt} - v_{ejk,t-1}$. As with the domestic revenue equation, we systematically examine this condition by constructing the accompanying export propensity equation. Modifying equation (15), based on the discussion in the preceding section, the number of "fresh" exporters, $n_{fjkt}$, who establish their firms after the export network commences in period $t_{ejk}$, divided by the number of potential entrepreneurs, $s_{jt}$, is specified as follows:

$$\begin{aligned}\frac{n_{fjkt}}{s_{jt}} = A_{jk} &+ \sum_{t'=t_{ejk}+1}^{t} \frac{q_{ekt'} - q_{dkt'}}{a(\delta-1)(1-\alpha)} + \theta_e \sum_{t'=t_{ejk}+1}^{t} \frac{\log n_{ejk,t'-1}}{a(\delta-1)(1-\alpha)} \\ &- \theta_d \sum_{t'=t_{ejk}+1}^{t} \frac{\log n_{jk,t'-1}}{a(\delta-1)(1-\alpha)} + \sum_{t'=t_{ejk}+1}^{t} \frac{v_{ejkt'}}{a(\delta-1)(1-\alpha)} \end{aligned} \quad (19)$$

There are five terms in this equation: a fixed birth county-destination prefecture term, a market-time term, an export network term, a domestic network overhang term, and an unobserved productivity term.[16] In contrast with the equations we have derived thus far, notice that the network terms include the history of relevant network sizes, rather than a single lag. However, if we double-difference equation (19), then we are left with the standard set of covariates: (i) destination-time period effects, (ii) $\log n_{ejk,t-1} - \log n_{ejk,t-2}$, and (iii) $\log n_{jk,t-1} - \log n_{jk,t-2}$, with $v_{ejk,t-1} - v_{jk,t-2}$ in the error term.

[16] Looking back at equation (15), the network terms are multiplied by cohort size, $s$. In the modified specification of the equation, this is replaced by $s_{jt}/a$, where $a$ is the number of (equally sized) age cohorts.

Recall that our analytical model made the simplifying assumption that the domestic network and the export network start at the same time. If this were true, then Proposition 1 implies that domestic producers never transition to (mixed) exporting. Once we extend the model to allow the export network to start at a later date than the domestic network, then high ability domestic producers who entered a prefecture before the export network was established; i.e. between periods $t_{djk}$ and $t_{ejk}$, will become mixed exporters after period $t_{ejk}$ if their export profits exceed $\beta$, as in Melitz (2003). This condition is more likely to be satisfied as the export network expands, with an accompanying increase in the productivity multiplier $C_{ejkt}$, bringing in more mixed exporters over time. Incorporating this extension to the model, the share of domestic producers from birth county $j$ who were active in prefecture $k$ when the export network was established and then later became mixed exporters is specified as follows:

$$\frac{n_{mjkt}}{n_{jkt_{ejk}}} = \frac{A - \log \omega^*_{mjkt}}{A - \omega^*_{djkt_{ejk}}} = \frac{A - \log \omega^*_{mjkt}}{n_{jkt_{ejk}}/s_{jt_{ejk}}}$$

Rearranging terms and replacing $A$ with $A_{jk}$,

$$\frac{n_{mjkt}}{s_{jt_{ejk}}} = A_{jk} + \frac{q_{ekt}}{\delta(1-\alpha)} + \theta_m \frac{\log n_{ejk,t-1}}{\delta(1-\alpha)} + \frac{v_{ejkt}}{\delta(1-\alpha)} \tag{20}$$

Henceforth, we will call these mixed exporting firms "incumbent exporters," to distinguish them from the "fresh exporters" described above, who enter after period $t_{ejk}$ and export immediately.[17] First-differencing the preceding equation to purge the fixed effects, we are left with destination-time period effects and $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ as covariates, with $v_{ejkt} - v_{ejk,t-1}$ in the error term.

Among the fresh exporters, some pure exporters will also later become mixed exporters, as in the analytical model, but their numbers will be determined by the size of the domestic network, which determines whether their profit from entering domestic production exceeds $\beta$. These firms are included in $n_{fjkt}$ on the left hand side of equation (19) and thus there is no double-counting: the stock of export firms, $n_{ejkt} = n_{fjkt} + n_{mjkt}$. $n_{fjkt}$ is positively correlated with $v_{ejkt}$ in equation (19) and $n_{mjkt}$ is positively correlated with $v_{ejkt}$

[17] To derive $\log \omega^*_{mjkt}$, we set the profit from exporting, as specified in equation (8), equal to $\beta$.

in equation (20). Lagging one period, it follows that $n_{ejk,t-1}$ is positively correlated with $v_{ejk,t-1}$. Following the same argument that we used for the preceding analysis of domestic production, the export network effects in equations (18)-(20) will be biased.

The first instrument that we construct for the growth in export network size, $\log n_{ejk,t-1} - \log n_{ejk,t-2}$, once equations (18) and (20) have been first-differenced and equation (19) has been double-differenced, is the export network duration, $t - t_{ejk}$. However, additional instruments are now available because the domestic network overhang term restricts entry into exporting in equation (19). In particular, any variable that determines the growth in domestic network size will, by extension, determine the change in fresh exporter entry in the double-differenced equation. The change in fresh exporter entry and the growth in export network size are mechanically correlated, although the sign of this relationship is theoretically ambiguous, and hence this variable will be a valid instrument. We can thus use TFP shocks in the traditional occupation in the birth county and domestic network duration, $t - t_{djk}$, as additional instruments for the growth in export network size.[18]

We close this section with a comment on the method that is used to purge fixed effects. As in the dynamic panel literature, we first-difference all the estimating equations, with the exception of the fresh exporter propensity equation where we double-difference. One reason why we do this is because the growth in network size then appears on the right hand side of the transformed estimating equations. This variable has a convenient interpretation in the context of the model, guiding the construction of suitable instruments. A second reason why we difference in this way is because the numerator on the left hand side of the propensity equations is transformed from the stock of firms to firm entry. This will allow us to complete the empirical analysis by quantifying the contribution of the birth county networks to firm entry.

[18]We allow $v_{djkt}$ and $v_{ejkt}$ to be correlated and, hence, these variables will also serve as instruments for the (endogenous) growth in domestic network size in the double-differenced equation (19). TFP shocks in the traditional occupation cannot be used as an instrument for the growth in domestic network size in equation (17) when it is first differenced because $u_{jkt} - u_{jk,t-1}$ appears in the error term. However, this is not the case in equation (19).

## 5. TESTING THE MODEL

### 5.1. *Identifying Network Effects*

When networks are active, a firm will perform better when more firms from its birth county are present in the same prefecture. As specified in equations (16) and (18), this implies that the firm's performance – revenue – will be increasing in the size of its network, conditional on destination-time period effects and firm fixed effects.

While the SAIC registration database provides the location and sector of each firm, it does not include information on its performance. To measure the performance of domestic producers, we thus turn to the SAIC inspection database, which provides revenues for a subset of registered firms over time.[19] To measure the size of the domestic network, we return to the SAIC registration database, which allows us to construct the (lagged) stock of firms from birth county $j$ located in prefecture $k$ in period $t-1$. Revenues for exporters are recorded, by shipment, in the Customs database and hence can be aggregated to the firm level. Network size for export firms can be constructed by merging the SAIC registration database with the Customs database.[20]

Although we include a rich set of covariates in the estimating equations, the estimated network effects could still be biased on account of the birth county-destination prefecture productivity shocks. In the previous section, we proposed two types of instruments to address these biases.

The first (shift-share) instrument that we construct is based on the idea that TFP shocks to non-business activities in the rural birth county such as agriculture and related wage labor will push potential entrepreneurs into (non-agricultural) business, independently of unobserved pull factors. Following Imbert et al. (2022), we construct the shift-share instrument in the following steps: (i) Using time series variation in world crop prices, and assuming that these prices follow an AR1 process, we construct a price shock in each year for 11

[19]The inspection database has reasonable sample size for 20 (out of 31) provinces from 1998 onwards and, hence, the sample that we use for the analysis spans the 1998-2012 period with this restricted set of provinces. It is possible that selection into this sample is non-random, but the firm fixed effects that we include in the estimating equations will account for any resulting bias.

[20]As noted, most export firms are also engaged in domestic production and, hence, they do not necessarily send shipments abroad in each year. We thus designate a firm as being an export firm in a given year if it has appeared in the Customs database in the past and continues to be active; i.e. it remains in the SAIC registration database.

major crops that account for 96 percent of cultivated area in China. (ii) For a given birth county, we weight each crop's price shock by a fixed factor that reflects its contribution to local agricultural production (by value) to construct a composite agricultural income shock in each year. (iii) We assume that the decision to establish a firm and, hence, firm entry in a given year is based on the average of the income shocks in that year and the preceding two years. (iv) The entering firms are then "distributed" across destination prefectures by dividing the county-level income shocks by distance (plus one). This adjustment is motivated by the standard gravity model, but does not require the estimation of any auxiliary parameters.[21] Further details of variable construction are provided in Appendix C.2.

The second instrument is simply the duration of each domestic or export network. This instrument can be constructed because our administrative data, covering the universe of registered firms, goes back to the 1980's, which is prior to the market reforms. It is thus possible to pin down the precise point in time at which each birth county-destination prefecture network commenced, typically with a single firm. As discussed in the preceding section, the time of commencement and, by extension, network duration is plausibly exogenous. The growth rate of the network also varies with its duration, both theoretically and empirically, thus satisfying the conditions for a valid instrument.

Table II reports first-stage estimates of the first-differenced revenue equations (16) and (18). The dependent variable in Column 1 is the growth in domestic network size: $\log n_{jk,t-1} - \log n_{jk,t-2}$. Note that first-differencing purges firm fixed effects, but the destination-time period effects are retained as covariates in the first-stage equation. We see in Column 1 that the coefficient on the birth county agricultural shocks is negative and significant, indicating that negative TFP shocks to the traditional occupation (the outside option) are pushing potential entrepreneurs into business, as implied by the model. The coefficient on the domestic network duration is also negative and significant.[22]

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[21] We could, in principle, estimate a gravity equation that would provide us with a "migration" elasticity with respect to distance between the birth county and any given destination prefecture. However, the dependent variable in the gravity equation is the share of firms from the birth county in each destination prefecture, which is endogenous to the unobserved productivity shocks. This implies that the estimated elasticity will also be correlated with these shocks. Our distance adjustment effectively sets the elasticity to be minus one, as described in Appendix C.2. This reduces the (predictive) power of the instrument, but maintains its validity.

[22] The negative coefficient is not inconsistent with Figure 1a, where the change in domestic network size is increasing with duration. The dependent variable in Table II is measured by the difference in log network size. If

TABLE II

FIRST-STAGE ESTIMATES OF THE (FIRST-DIFFERENCED) FIRM REVENUE EQUATIONS

| | (1) difference in log domestic network size | (2) difference in log export network size |
|---|---|---|
| Birth county agricultural shocks | -0.219** | -0.203 |
| | (0.105) | (0.113) |
| Domestic network duration | -0.004*** | 0.004*** |
| | (0.000) | (0.000) |
| Export network duration | – | -0.014*** |
| | | (0.002) |
| Destination-time period effects | Yes | Yes |
| Observations | 5,065,050 | 281,091 |

Note: network size is constructed from SAIC registration data and Customs data.
The dependent variable in the first-differenced first-stage equation is measured by $\log n_{jk,t-1} - \log n_{jk,t-2}$ for the domestic network and $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ for the export network.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

Turning to Column 2, the dependent variable is now the growth in export network size; $\log n_{ejk,t-1} - \log n_{ejk,t-2}$, and we see that the coefficient on export network duration is negative and significant, matching the sign of the coefficient on domestic network duration in Column 1. As discussed in the preceding section, any variable that is a valid instrument for the growth in domestic network size will also be a valid instrument for the growth in export network size if the domestic network overhang is present. While the coefficient on domestic network duration is very precisely estimated in Column 2, the coefficient on the birth county agricultural shocks is statistically insignificant at conventional levels. We will thus use export network duration and domestic network duration as instruments for the growth in export network size in the analysis that follows. Although we do not have a theoretical prediction for the sign of the domestic network duration effect in Column 2, the fact that it switches from Column 1 tells us that this effect is working through different channels in the two equations.

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we measured the difference in levels rather than logs, then the association with network duration would be positive and significant, as reported in Appendix Table C1. This is also true for export network duration, which we discuss below.

TABLE III

SECOND-STAGE ESTIMATES OF THE FIRM REVENUE EQUATIONS

| Estimation: | OLS | | 2SLS | |
|---|---|---|---|---|
| | (1) log domestic revenue | (2) log export revenue | (3) log domestic revenue | (4) log export revenue |
| Log network size | 0.474*** | 0.526*** | 1.333*** | 0.933*** |
| | (0.010) | (0.019) | (0.088) | (0.125) |
| Firm fixed effects | Yes | Yes | Yes | Yes |
| Destination-time period effects | Yes | Yes | Yes | Yes |
| Kleibergen-Paap F | – | – | 95.52 | 26.91 |
| Hansen J | – | – | 8.46 | 74.13 |
| Observations | 5,065,050 | 281,091 | 5,065,050 | 281,091 |

Note: network size is constructed from SAIC registration data and Customs data.
Revenue is constructed from SAIC inspection data and Customs data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variable is thus measured by the growth in its size: $\log n_{jk,t-1} - \log n_{jk,t-2}$ for the domestic network and $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ for the export network.
Instruments for the growth in domestic network size: birth county agricultural shocks, domestic network duration.
Instruments for the growth in export network size: export network duration, domestic network duration.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

Table III reports the revenue equations (16) and (18), with OLS estimates in Columns 1-2 and 2SLS estimates in Columns 3-4.[23] The network size effect is positive and significant without exception, with the point estimates increasing in magnitude when we instrument for network size. Recall that first-differencing to purge the fixed effects will bias the estimated network effect downward, whereas (positive) serial correlation in the unobserved productivity shocks at the destination will generate bias in the opposite direction. The observed bias, which is negative on net, indicates that the first mechanism dominates in this setting. While the Kleibergen-Paap F statistic indicates that the instruments have sufficient power, notice that we reject the Hansen J over-identification test with both outcomes (the 5 percent critical value is 3.84). This implies that the individual instruments have heterogeneous treatment effects, and this heterogeneity will help us interpret some of the results reported below.

[23] We discard the top one percentile and the bottom one percentile of the first-differenced dependent variable when estimating all of the firm revenue equations to remove outliers.

### 5.2. *Validating the Statistical Instruments*

We used two types of instruments to identify network effects above: a shift-share instrument, leveraging exogenous agricultural productivity shocks in the birth county, and network durations, which are determined by the exogenous timing of initial entry, separately for domestic production and exporting. In this section, we provide additional support for the validity of these instruments.

We begin by estimating the revenue equations, for domestic production and exporting, separately with each instrument in Table IV. If treatment effects are heterogeneous, as we expect based on the tests of the overidentifying restrictions in Table III, then the excluded instruments will have a direct effect on the outcome and thus belong in the estimating equation (Mogstad et al., 2021). The coefficients on the excluded instruments are indeed statistically significant in Table IV, with one exception. Nevertheless, the estimated network effects are positive and significant with each individual instrument, for both outcomes, increasing our confidence in their validity. The discussion that follows will subject each instrument to further scrutiny.

**Shift-Share Instrument**: We assess whether the instrument based on agricultural shocks in the birth county satisfies the exclusion restriction, using the estimates in Table IV, Column 1 as the reference point. In recent years, shift-share instruments have received much attention in the economics literature. We follow this literature and implement a series of robustness tests in Appendix C.3 that validate each component of the shift-share instrument.

**1. Agricultural price shocks**: The firms in our sample are engaged in (non-agricultural) manufacturing and services. It follows that their revenues will not be directly impacted by agricultural price shocks. However, one way in which these shocks (interacted with birth county crop shares) could indirectly determine firm revenues is if they affect the local economy through general equilibrium effects and the firm is located in the birth county itself. A second way in which agricultural price shocks could affect a firm's revenues is if it is operating in the food processing sector. This is true regardless of where the firm is located. Finally, a third way in which agricultural price shocks could directly affect business is through the wealth channel. If own (family) wealth is used to finance business, as in Song et al. (2011), then a negative price shock will curtail the operations of entrepreneurs from

TABLE IV

SECOND-STAGE ESTIMATES OF THE FIRM REVENUE EQUATIONS BY INSTRUMENT

| Dependent variable: | log domestic revenue | | log export revenue | |
|---|---|---|---|---|
| Instrument: | birth county agricultural shock | domestic network duration | export network duration | domestic network duration |
| | (1) | (2) | (3) | (4) |
| Log network size | 2.276*** | 1.277*** | 0.778*** | 1.810*** |
| | (0.696) | (0.086) | (0.129) | (0.188) |
| Domestic network duration | 0.004 | – | 0.004*** | – |
| | (0.003) | | (0.000) | |
| Birth county agricultural shocks | – | -0.219*** | – | – |
| | | (0.073) | | |
| Export network duration | – | – | – | 0.014*** |
| | | | | (0.002) |
| Firm fixed effects | Yes | Yes | Yes | Yes |
| Destination-time period effects | Yes | Yes | Yes | Yes |
| Kleibergen-Paap F | 4.35 | 187.69 | 54.51 | 65.13 |
| Observations | 5,065,050 | 5,065,050 | 281,616 | 281,616 |

Note: network size is constructed from SAIC registration data and Customs data.
Revenue is constructed from SAIC inspection data and Customs data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variable is thus measured by the growth in its size.
Instruments for the growth in domestic network size: birth county agricultural shocks or domestic network duration.
Instruments for the growth in export network size: export network duration or domestic network duration.
The excluded instrument is included as a covariate in each case.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

agricultural families. This is true regardless of the location in which they are active and will result in a decline in their revenues. Appendix Table C2 shows that the results are retained when (i) the analysis is restricted to firms operating outside the birth county, (ii) when firms operating in food processing and related sectors are removed from the sample, and (iii) when the agricultural income shocks are included directly in the estimating equation to account for the wealth effect.

**2. Crop shares**: While the direct effect of the crop shares is subsumed in the firm fixed effects, we consider the possibility that the interaction of these shares with time determines firm revenues. As Goldsmith-Pinkham et al. (2020) note, these interaction effects must also be considered when examining the validity of any shift-share instrument.[24] To do this,

[24]For example, suppose that (historical) cultivation of a particular crop is associated with an entrepreneurial culture or a greater willingness to bear risk in the local population. If these traits have a differential effect on

we build on Goldsmith-Pinkham et al's insight that if the crop shares are exogenous, then the shift-share instrument that we construct is "equivalent" to using the shares associated with each crop, interacted with time effects, as independent instruments for network size. We first verify in Appendix Table C3 that the estimated network effect, with crop shares interacted with time effects as instruments, is similar to the benchmark estimate in Table IV, Column 1. We then proceed to estimate the network effect separately with each crop share interacted with the time effects as the instrument. These instrumental variable estimates could vary across crops if treatment effects are heterogeneous. As Goldsmith-Pinkham et al. note, however, such heterogeneity may be a concern if it is substantial or if any crop is an outlier. We see in Appendix Table C3 that the network effects estimated with each crop are positive, statistically significant, and similar in magnitude to each other.

**3. Distance adjustment**: Distance is a fixed characteristic and, hence, its direct effect on firm revenue is subsumed in the firm fixed effect. However, its interaction with time must also be considered when examining the validity of the shift-share instrument, as above. Suppose that firms located at a greater distance from their rural origin are established in faster-growing prefectures. Distance interacted with time will then determine firm revenue, but this does not undermine our identification strategy because destination-time period effects are included in the estimating equation. The threat to identification with this component of the shift-share instrument is that particular types of individuals may choose to move far away and the outcomes for those types may vary differentially with experience or at different stages of economic development. To address the preceding concern, we include distance interacted with time effects in the estimating equation (see Appendix Table C2) and verify that the estimated network effects are unchanged.

**Network Duration Instruments**: Our primary argument supporting the validity of the network duration instrument is based on the observation that the prototypical birth county-destination prefecture network commences with a single individual. If initial entry was generated by a community level factor, which, in turn, was correlated with the unobserved

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firm revenue over time with economic development, then our instrument would violate the exclusion restriction. Alternatively, if counties growing particular crops industrialize relatively fast due to the nature of the agricultural production technology, then entrepreneurs born in those counties will have preferred access to capital (to the extent that firms are self-financing). This would undermine the validity of the instrument once again.

productivity shock $v_{bjkt}$, $b\{d, e\}$, then for any reasonable specification of the ability distribution, we would have expected to see a positive mass of entrants instead.

With export revenue as the outcome, *domestic* network duration is a particularly credible instrument because it determines *export* network size through the indirect domestic network overhang channel. The domestic network is established long before the export network; in our data, this network starts, on average, 11 years earlier than the export network in those birth county-prefectures where both networks co-exist. The conditions that gave rise to the domestic network are unlikely to be related to those that start the export network. The fact that domestic production and exporting are different activities, with distinct networks (as verified below), gives us additional confidence in the validity of the domestic network duration instrument.

Entrepreneurial experience will be mechanically correlated with network duration. Although experience effects do not feature in the model, they could potentially determine firm outcomes directly, undermining the domestic and export network duration instruments in the domestic revenue and export revenue equations, respectively. We address this possibility in Appendix Table C4 by including entrepreneurial experience, measured as the time since the firm was established, as an additional covariate in the first-differenced revenue equations. The estimated network effects continue to be positive and significant, both for domestic production and exporting.

We complete this section by comparing and contrasting the duration instruments with the identification strategy that is conventionally adopted in the dynamic panel literature. The canonical dynamic panel model includes the lagged dependent variable and a fixed effect as covariates in the estimating equation. When this equation is first-differenced, it induces a correlation between the growth in the lagged dependent variable on the right hand side and the residual term, exactly as described in Section 4.1. If the error term is serially uncorrelated, then the twice lagged dependent variable can be used as an instrument. If the error term is serially correlated, but dampens sufficiently fast, then lags further back in time can be used as instruments (Arellano and Bond, 1991, Blundell and Bond, 1998). However, the lags that are used will typically be recent because most panel datasets are short and because longer lags of the dependent variable will lack predictive power. Our estimating equations, particularly the propensity equations that we estimate in the section that follows, are closely related to the canonical dynamic panel equation and face the same

source of bias. However, a unique feature of our data is that we observe the precise point in time when the dynamic process commences in each birth county-destination prefecture. We can thus exploit the network multiplier effect to construct an alternative, more robust, instrument that is based on exogenous entry shocks that determine the network's duration, rather than on recent lagged values of the dependent variable.

### 5.3. *Cross-Network Effects*

The model specifies that domestic producers benefit from a network consisting of all firms in the birth county-destination prefecture, whereas export revenues are determined by a restricted network consisting of export firms alone. Suppose, instead, that export revenues are in addition declining in the size of the domestic network, perhaps because it crowds out inputs and connections. Then this would provide an alternative explanation for the negative effect of domestic network size on export entry that does not rely on the overhang channel.

To verify that cross-network effects are absent, as assumed in the model, the estimating equations in Table V include both domestic network size and export network size. The OLS estimates in Columns 1-2 are consistent with the model: domestic revenues are determined exclusively by the domestic network and export revenues are determined by the export network alone. The instrumental variable estimates with domestic revenue as the outcome in Column 3 are broadly consistent with the corresponding OLS estimates in Column 1. However, the OLS and 2SLS estimates in Columns 2 and 4, with export revenue as the outcome, are very different. The coefficient on domestic network size, in particular, becomes negative, large (in absolute magnitude), and statistically significant when we instrument for network sizes.

The standard interpretation when OLS and 2SLS estimates differ substantially is that there is a weak instrument problem, but this is not the case here because the Kleibergen-Paap F statistic is above 25. Our explanation for the divergence in Columns 2 and 4 is that the OLS and 2SLS estimates of the export revenue equation are not comparable, on account of the heterogeneous treatment effects we have uncovered and the domestic network overhang that is implied by our model. Notice that the specification in Table IV, Column 3 is the same as the specification in Table V, Column 4, except that domestic network duration, which appears as a covariate with a positive coefficient in the former table is replaced by

TABLE V

ESTIMATES OF THE FIRM REVENUE EQUATIONS: ALLOWING FOR CROSS-NETWORK EFFECTS

| Estimation: | OLS | | 2SLS | |
|---|---|---|---|---|
| | (1) log domestic revenue | (2) log export revenue | (3) log domestic revenue | (4) log export revenue |
| Log domestic network size | 0.527*** | 0.065 | 1.432*** | -1.139*** |
| | (0.030) | (0.050) | (0.094) | (0.129) |
| Log export network size | 0.008 | 0.521*** | -0.049 | 1.184*** |
| | (0.005) | (0.021) | (0.072) | (0.131) |
| Firm fixed effects | Yes | Yes | Yes | Yes |
| Destination-time period effects | Yes | Yes | Yes | Yes |
| Kleibergen-Paap F | – | – | 11.47 | 28.60 |
| Observations | 3,234,611 | 281,616 | 3,234,611 | 281,616 |

Note: network size is constructed from SAIC registration data and Customs data.
Revenue is constructed from SAIC inspection data and Customs data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variables are thus measured by the growth in their size.
Instruments for the growth in domestic and export network size in Columns 3-4: birth county agricultural shocks, domestic network duration, export network duration.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

(instrumented) domestic network size in the current table. We know from Table II, Column 1 that the association between domestic network size and domestic network duration is negative, once we first-difference the estimating equation. This result, together with the positive coefficient on domestic network duration in Table IV, explains why the sign of the coefficient on domestic network size is now negative. Making this argument more precise, the product of the first-stage coefficient in Table II, -0.004, and the domestic network size coefficient in Table V, -1.139, is equal to the "reduced form" domestic duration coefficient in Table IV, which is 0.004.

The intuition underlying the preceding explanation is that once we include domestic network size in the estimating equation, the coefficient on export network size can be interpreted as a local average treatment effect (associated with the export network duration instrument). Since treatment effects are heterogenous, this leaves room for the domestic network duration instrument to affect the export revenue through its independent effect on export network size. An alternative interpretation of the estimates in Column 4 is that a larger domestic network directly reduces export revenues, as discussed above. However, if

this were true, then we would also expect to observe a negative and significant coefficient on domestic network size with the OLS estimates in Column 2. It is difficult to credibly argue that bias in these estimates would exactly offset the "true" negative coefficient. Notice also that this is the only specification in Tables III and V where the OLS and 2SLS estimates diverge so substantially, and heterogeneous treatment effects together with the domestic network overhang provide a simple explanation for this divergence.

## 6. QUANTITATIVE ANALYSIS

### 6.1. *Firm Entry*

The evidence presented thus far indicates that domestic networks increase domestic revenues and export networks increase export revenues. To quantify the contribution of the networks to firm entry, we next proceed to estimate the structural propensity equations.

Recall from Section 4 that we are left with destination-time period effects and the growth in domestic network size when we first-difference the entrepreneurial propensity equation (17) to purge birth county-destination prefecture effects. Domestic network duration can be used as an instrument for the growth in domestic network size in the first-differenced equation. However, the birth county agricultural shocks, which were also used as instruments in the domestic revenue equation, are no longer valid since they directly determine entry into business.

As with the export revenue equation, we use export network duration and domestic network duration (via the overhang effect) as instruments for the growth in export network size in the first-differenced incumbent exporter propensity equation (20). For the fresh exporter equation (19), we need to double-difference to arrive at the standard set of covariates. Birth county agricultural shocks, domestic network duration, and export network duration can then be used as instruments for the growth in domestic network size and the growth in export network size.[25]

The propensity measures on the left hand side of equations (17), (20), and (19) are constructed by dividing the relevant stock of firms by the number of potential entrepreneurs,

[25]When we first-difference, we are left with firm entry as the dependent variable. This variable is (almost) always zero in networks that have little or no growth after inception. Since the double-difference is not meaningful in that case, we drop all networks that had zero entry or in which just one firm entered (at some arbitrary point in time) when estimating the fresh exporter propensity equation.

$s_{jt}$, at time $t$. As discussed in Section 4.1, $s_{jt}$ is measured by the number of individuals in the $[\underline{a}, \overline{a} - 1]$ age range in period $t$ who were born in county $j$. In practice, we set $\underline{a}$ equal to 25 and $\overline{a}$ equal to 45, using the 1990 population census to construct $s_{jt}$ (see Appendix D.1 for details). When first-differencing a propensity equation, we replace $s_{jt}$, $s_{j,t-1}$ in the denominator on the left hand side by their average, which is reasonable since the number of potential entrepreneurs is a slowly evolving variable. The advantage of this approximation is that the numerator on the left hand side of the first-differenced equation then becomes the number of entering firms. For the fresh exporters, we will need to double-difference the propensity equation and, hence, the model will predict the change in entry at each point in time.

We see that the coefficient on domestic network size in Table VI, Column 1, with entrepreneurial propensity as the dependent variable and the coefficient on export network size in Column 2, with incumbent exporter propensity as the dependent variable, are both positive and significant.[26] As implied by the model, and in line with the evidence in Table II and Table V that is indicative of a domestic network overhang, we see that the coefficient on the export network term is positive and significant, while the coefficient on the domestic network term is negative and significant in Table VI, Column 3. The negative coefficient on the domestic network term reduces the number of fresh exporters, who account for 70 percent of all exporters in our data, and we will quantify this effect in the section that follows.

### 6.2. *Decomposition Analysis*

Based on the first-differenced (or double-differenced) propensity equations, the determinants of firm entry in our model can be partitioned into two components: an exogenous component, which includes the destination-time period effects and all other unobserved factors (the error term) and an endogenous network component.[27] Our instrumental variable estimates of the network effects allow us to quantify the contribution of the network

[26] When estimating all of the entry equations, we discard the top one percentile and the bottom one percentile of the first-differenced dependent variable to remove outliers.

[27] The dependent variable in the double-differenced fresh exporter propensity equation is the change in entry. However, we can recover the level of entry at each point in time in each network by solving recursively.

TABLE VI

PROPENSITY EQUATION ESTIMATES

| | (1) entrepreneurial propensity | (2) incumbent exporter propensity | (3) fresh exporter propensity |
|---|---|---|---|
| Log network size | 0.252*** | 0.002*** | – |
| | (0.012) | (0.000) | |
| Summed log export network size | – | – | 0.002*** |
| | | | (0.001) |
| Summed log domestic network size | – | – | -0.004*** |
| | | | (0.001) |
| Birth county-destination fixed effects | Yes | Yes | Yes |
| Destination-time period effects | Yes | Yes | Yes |
| Kleibergen-Paap F | 359.71 | 321.37 | 60.87 |
| Observations | 1,043,470 | 20,265 | 11,520 |

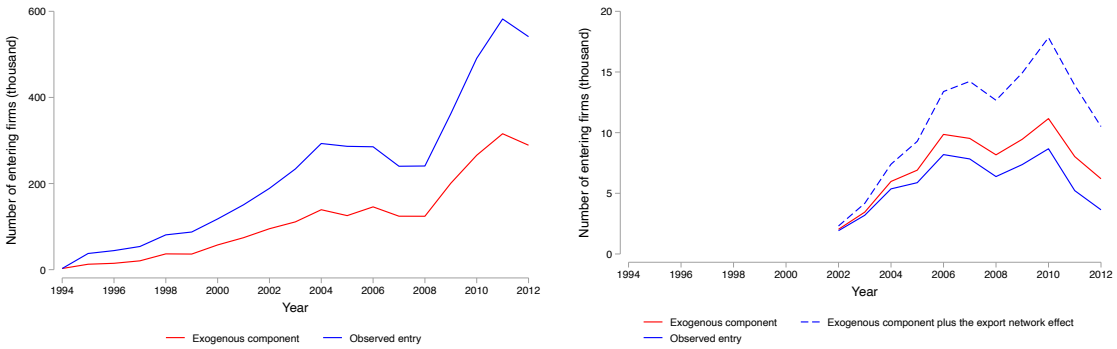
Note: propensity and network size, measured at the birth county-destination prefecture-time period level, are constructed from SAIC registration data and Customs data.
Network size is measured by the lagged stock of all firms with entrepreneurial propensity as the dependent variable and by the lagged stock of export firms with incumbent exporter propensity as the dependent variable.
Summed network sizes are computed over the network's history with fresh exporter propensity as the dependent variable.
Birth county-destination prefecture fixed effects are purged by first-differencing prior to estimation and the modified network variables in columns 1-2 are thus measured by the growth in their size.
Instrument for $\log n_{jk,t-1} - \log n_{jk,t-2}$ in column 1: domestic network duration.
Instruments for $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ in column 2: export network duration, domestic network duration.
The fresh exporter propensity equation is double-differenced to once again measure the modified network variables by the growth in their size.
Instruments for $\log n_{jk,t-1} - \log n_{jk,t-2}$ and $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ in column 3: domestic network duration, birth county agricultural shocks, export network duration.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

component to firm entry at each point in time. This, in turn, allows us to back out the contribution of the (residual) exogenous component to firm entry.

The red line in Figure 2a plots the contribution of the exogenous factors to firm entry. As can be seen, this contribution is increasing over time. The blue line in the figure plots the observed entry of firms in each year. The gap between the two lines measures the contribution of the networks, and this is also increasing over time. Based on our estimates, the birth county networks account for about half of the observed entry in 2012.

We can implement a similar decomposition exercise with the first-differenced domestic revenue equation to quantify the contribution of the birth county networks to that outcome.

Based on our estimates, if the networks were absent in 2012, then average firm revenues in that year would have been 22 percent smaller.[28] This effect is of a similar order of magnitude as the entry effects that we have estimated, which is as expected since entry is determined by revenues in our model. The revenue estimates have been subjected to a number of robustness tests and, hence, this consistency gives us additional confidence in the magnitude of the estimated entry effects.




(a) All Firms (b) Exporters

FIGURE 2.—Decomposition Analysis

Source: SAIC registration data and Customs data.

The red line in Figure 2b plots the contribution of the exogenous factors to the entry of export firms, combining incumbent exporters and fresh exporters. As can be seen, this contribution is increasing over time, presumably due to improved export opportunities and increasing government support. The gap between the red line and the dashed blue line measures the contribution of the export networks to entry, once again combining the incumbent exporters and fresh exporters. Based on our estimates, the export networks doubled the entry of export firms relative to the counterfactual (red line) in which networks were active. They thus had a similar proportional impact on firm entry as the domestic networks in Figure 2a.

However, there is an additional component to export entry, which is the domestic network overhang. The contribution of this term to overall export entry is measured by the gap

[28] This quantification exercise is implemented in two steps: First, we compute the effect of removing the network on the dependent variable in the first-differenced revenue equation. Second, we back out the resulting effect on current revenue, treating lagged revenue as predetermined.

between the dashed blue line and the solid blue line (observed entry) in Figure 2b. As can be seen, this gap is increasing over time, matching the increase in domestic network size that is implied by Figure 2a. The negative domestic network effect more than offsets the positive contribution of the export network, resulting in a decline in the number of entering exporters relative to the counterfactual without networks. While the rural-born entrepreneurs were able to exploit their networks to take advantage of the economic opportunities that became available in the initial stage of economic development, the birth county networks had a net negative effect on entry in the second stage.

## 7. CONCLUSION

This research provides a network-based explanation for the influx of rural-born entrepreneurs in China, following the market reforms of the early 1990's. This explanation is based on the idea that birth county networks provided mutual support to their members, filling the gap in an environment where many markets were missing or incomplete. Our estimates indicate that the number of entering rural-born entrepreneurs in 2012, the end point of our analysis, would have been halved in the absence of these networks. Although the existence of community-based business networks has been documented in historical and in contemporary industry studies, this constitutes the first economy-wide evidence to date of the critical role played by these informal institutions.

While the domestic networks that we identify may have facilitated mobility in the initial transition, our model implies that they could dampen the growth of newly emerging export networks and, as a result, the transition to the next stage of economic development. Indeed, our estimates indicate that the increase in the number of entering exporters that was generated by the export networks was more than offset by the domestic network overhang. This resulted in a decline in the entry of export firms relative to the counterfactual without networks in the rural-born population that we study .

Self-interested entrepreneurs do not internalize the effect of their entry on the performance of other firms (through the change in network size) and, hence, there is a role for policy. Export subsidies, which have no consequence for domestic profits, are unambiguously efficiency enhancing. In contrast, entry subsidies must balance two opposing effects: their positive effect on domestic profits due to a larger domestic network and the negative effect on export profits due to a smaller export network (on account of the domestic net-

work overhang). If the latter effect is sufficiently large, it may even be optimal to tax entry. Adding to the complexity, if the second transition is anticipated, domestic policies during the first transition would need to take account of their future consequences for exporting. Although a complete characterization of dynamic optimal subsidies is left to future research, we note that industrial policy could have large positive impacts in economies where networks are active.

The organic process of economic development that we describe in this paper, in which networks emerge at each stage to facilitate the occupational mobility of their members, and pre-existing networks slow down the growth of networks at subsequent stages, may not be specific to China or to business activities. At the same time, the analysis in this paper will only be relevant in populations where community networks are already active or have the potential to be activated and this will, in general, depend on the underlying social structure. A natural question to ask is which social structures are more conducive to network formation and growth. One way to answer this question would be to examine heterogeneity in social structure across populations, with its consequences for network performance and occupational mobility. Such heterogeneity would have further implications for industrial policy, with optimal subsidies now varying across space. These extensions to the current analysis are also left to future research.

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## APPENDIX A: NETWORK CHARACTERISTIC

TABLE A1

INDUSTRY COMPOSITION WITHIN BIRTH COUNTY-DESTINATION PREFECTURE-TIME PERIODS

| | Domestic network | Export network |
|---|---|---|
| | (1) | (2) |
| Percent of firms in the most popular industry | 66.9 | 79.6 |
| Percent firms in related industries | 26.4 | 16.5 |

Note: statistics are computed using SAIC registration data and Customs data.
The network is defined at the birth county - destination prefecture level and is assumed to commence if we observe at least one firm for at least three years.
The domestic network includes all firms in the birth county-destination prefecture and the export network is restricted to export firms.
The most popular industry is defined as the 3-digit IO code that has the largest share of firms, for a given birth county-destination prefecture-time period.
Related industries are upstream-downstream or share complementary inputs-outputs with the most popular industry, as in Fan and Lang (2000).

TABLE A2

EVOLUTION OF BIRTH COUNTY-DESTINATION PREFECTURE NETWORKS OVER TIME

| | Domestic network | Export network |
|---|---|---|
| | (1) | (2) |
| Total number of networks | 166,117 | 11,365 |
| Percent of networks that are multi-firm | 71.2 | 44.7 |
| Percent of firms in multi-firm networks | 99.1 | 92.8 |
| Percent of multi-firm networks that start with one firm | 84.6 | 76.2 |

Note: statistics are computed using SAIC registration data and Customs data.
The network is defined at the birth county - destination prefecture level and is assumed to commence if we observe at least one firm for at least three years.
Networks that start with a single firm and stay that way are retained in the sample.
The domestic network includes all firms in the birth county-destination prefecture and the export network is restricted to export firms.

APPENDIX B: THE MODEL

## 1. Proposition 1

The following parameter restrictions ensure existence of a unique equilibrium with a positive mass of agents in each occupation at each date for every cohort:

$$\log\zeta > (A-1)(1-\sigma) + \frac{\theta_d \log(sT)}{1-\alpha} \tag{21}$$

$$\log\gamma > \frac{(\delta-1)\log\zeta}{1-\sigma} + \frac{\theta_e \log(sT)}{1-\alpha} \tag{22}$$

$$\log\beta > \frac{\log\gamma}{\delta-1} - \log\zeta + \frac{\theta_d \log(sT)}{1-\alpha} \tag{23}$$

$$A > \log\beta + \log\zeta \tag{24}$$

*Proof of Proposition 1*:

To prove the Proposition, we show that ability thresholds are interior and ordered given the parameter restrictions (21-24).

We begin by showing that $\log\omega^*_{dt} > A-1$ if (21) is satisfied. From (10):

$$\log\omega^*_{dt} = \frac{\log\zeta}{1-\sigma} - \frac{\log C_{dt}}{(1-\alpha)(1-\sigma)}$$

Observe that $sT$ is an upper bound on network size. Hence, $\theta_d \log(sT)$ is an upper bound on the network effect in the domestic market. It follows that (21) is a sufficient condition for $\log\omega^*_{dt} > A-1$.

Next, we show that $\log\omega^*_{et'} > \log\omega^*_{dt}$, for all $t \geq t'$, if (22) is satisfied. From (11):

$$\log\omega^*_{et'} = \frac{1}{\delta-1}\left[\log\gamma + \frac{\log C_{dt'} - \log C_{et'}}{1-\alpha}\right]$$

It follows that $\log\omega^*_{et'} > \log\omega^*_{dt}$ if

$$\log\gamma > \frac{(\delta-1)\log\zeta}{1-\sigma} - \frac{(\delta-1)\log C_{dt}}{(1-\sigma)(1-\alpha)} - \frac{\log C_{dt'} - \log C_{et'}}{1-\alpha}$$

$\log C_{dt}$, $\log C_{dt'}$ are non-negative, while $\log C_{et'}$ is bounded above by $\theta_e \log(sT)$. It follows that (22) is a sufficient condition for the preceding inequality to be satisfied.

Next, we show that $\log\omega^*_{mt} > \log\omega^*_{et'}$, for any $t \geq t'$, if (23) is satisfied. From (12):

$$\log\omega^*_{mt} = \log\beta + \log\zeta - \frac{\log C_{dt}}{1-\alpha}$$

It follows that $\log \omega^*_{mt} > \log \omega^*_{et'}$ if

$$\log \beta > \frac{\log \gamma}{\delta - 1} - \log \zeta + \frac{\log C_{dt}}{1 - \alpha} + \frac{\log C_{dt'} - \log C_{et'}}{(\delta - 1)(1 - \alpha)}$$

Since $\log C_{dt}$, $\log C_{dt'}$ are bounded above by $\theta_d \log(sT)$ and $\log C_{et'}$ is nonnegative, (23) ensures that the preceding inequality is satisfied.

Finally, we show that $\log \omega^*_{mt} < A$ for any $t \geq t'$, if (24) is satisfied. From the expression for $\log \omega^*_{mt}$ derived above, we require that

$$A > \log \beta + \log \zeta - \frac{\log C_{dt}}{1 - \alpha}$$

Since $\log C_{dt}$ is nonnegative, (24) ensures that the preceding inequality is satisfied. This concludes the proof.

**2. Extending the model to allow for forward looking behavior**

We now explain how our model extends to the case where agents are non-myopic, and apply a discount factor $\phi \in (0,1)$ to future profits. We show that expressions for optimal capital stocks and profits at any date (conditional on entry into any market) are unchanged. Moreover, the entrepreneurial propensity equation is unchanged for small values of $\phi$. The same is not true in general for the export propensity, for which a closed form expression can no longer be obtained, but (a) the expression for the case of myopic agents is an approximation for the case of small $\phi$ and (b) forward looking behavior is likely to induce an additional source of the domestic network overhang effect.

Suppressing notation for market and network sizes at different dates, the dynamic optimization decision faced by an agent of ability $\omega$ at date $t$ with inherited capital stocks $K_{d,t-1}, K_{e,t-1}$ is represented by the following Bellman equations. If the agent is a mixed exporter at $t-1$, i.e., $K_{d,t-1}K_{e,t-1} > 0$:

$$W_{mt}(\omega; K_{d,t-1}, K_{e,t-1}) = \max_{K_{dt} \geq K_{d,t-1}, K_{et} \geq K_{e,t-1}} [\pi_{dt}(\omega; K_{dt}) + \pi_{et}(\omega; K_{et}) - \beta + \phi W_{m,t+1}(\omega; K_{dt}, K_{et})] \tag{25}$$

where $\pi_{dt}(\omega; K_{dt}) \equiv C_{dt}\omega^{1-\alpha}K_{dt}^{\alpha} - rK_{dt}$ and $\pi_{et}(\omega; K_{dt}) \equiv C_{et}\omega^{\delta(1-\alpha)}K_{et}^{\alpha} - r(1+I)K_{dt}$.

If the agent is a pure exporter at $t-1$ (i.e., $K_{e,t-1} > 0, K_{d,t-1} = 0$):

$$\begin{aligned} W_{et}(\omega; K_{e,t-1}) = \max_{K_{dt} \geq 0, K_{et} \geq K_{e,t-1}} & [\pi_{et}(\omega; K_{et}) + \mathcal{I}_{K_{dt}>0}[\pi_{dt}(\omega; K_{dt}) - \beta \\ & + \phi W_{m,t+1}(\omega; K_{dt}, K_{et})] + (1 - \mathcal{I}_{K_{dt}>0})\phi W_{e,t+1}(\omega; K_{et})] \end{aligned} \tag{26}$$

where $\mathcal{I}_x$ is an indicator function taking value one if event $x$ happens and 0 otherwise.

If the agent is a pure domestic producer at $t-1$ (i.e., $K_{d,t-1} > 0, K_{e,t-1} = 0$):

$$\begin{aligned} W_{dt}(\omega; K_{d,t-1}) = \max_{K_{et} \geq 0, K_{dt} \geq K_{d,t-1}} & [\pi_{dt}(\omega; K_{dt}) + \mathcal{I}_{K_{et}>0}[\pi_{et}(\omega; K_{et}) - \beta \\ & + \phi W_{m,t+1}(\omega; K_{d,t-1}, K_{e,t-1})] + (1 - \mathcal{I}_{K_{et}>0})\phi W_{d,t+1}(\omega; K_{dt})] \end{aligned} \tag{27}$$

and finally if the agent has not already entered either market at $t-1$ (i.e., $K_{d,t-1} = K_{e,t-1} = 0$):

$$W_{ot}(\omega) = \max\{\omega^{\sigma} + \phi W_{o,t+1}(\omega); W_{dt}(\omega, 0); W_{et}(\omega, 0); W_{mt}(\omega; 0, 0)\} \tag{28}$$

Observe first that it continues to be the case that capital irreversibility constraints do not bind on the intensive margin, i.e., conditional on entering either domestic or export market, the associated optimal capital stocks are myopically optimal (e.g., $K_{dt}^{*}(\omega; K_{d,t-1}, K_{e,t-1})$) maximizes $\pi_{dt}(\omega; K_{dt})$ without any irreversibility constraint. The same proof applies: if we consider the relaxed problem where the irreversibility constraint is dropped, the constraint

does not bind since market and network sizes are growing. Hence the solution to the relaxed problem is a solution to the true problem. And in the relaxed problem, current capital stock (conditional on being positive) does not affect future profits, so it must be myopically optimal.

This implies that the value functions reduce to the following simpler expressions:

$$\begin{aligned} W_{mt}(\omega) &= \Pi_{Dt}(\omega) + \Pi_{Et}(\omega) - \beta + \phi W_{m,t+1}(\omega) \\ W_{et}(\omega) &= \max\{\Pi_{Et}(\omega) + \phi W_{e,t+1}(\omega); W_{mt}(\omega)\} \\ W_{dt}(\omega) &= \max\{\Pi_{Dt}(\omega) + \phi W_{d,t+1}(\omega); W_{mt}(\omega)\} \\ W_{ot}(\omega) &= \max\{\omega^{\sigma} + \phi W_{o,t+1}(\omega); W_{dt}(\omega); W_{et}(\omega); W_{mt}(\omega)\} \end{aligned} \tag{29}$$

where $\Pi_{Dt}, \Pi_{Et}$ denote static profits at date $t$ associated with myopically (unconstrained) optimal capital stocks provided in the text.

If all parameters lie in a compact set, these value functions are bounded and uniformly continuous. Hence for $\phi$ in a neighborhood of $0$, these value functions are close to those corresponding to $\phi = 0$, implying that the pattern of sorting will be similar, with ability thresholds for different options ordered as in the case of myopic agents (given in Proposition 1 of the text).

**Claim:** *For $\phi$ in a right neighborhood of $0$, the ability threshold $\omega^*_{dt}$ for entry into the domestic sector is the same as when agents are myopic ($\phi = 0$).*

The reasoning is as follows. As the pattern of sorting for small $\phi$ is similar to that where $\phi = 0$, the threshold $\omega^*_{dt}$ is determined by indifference between staying in the traditional occupation $o$ and entering the domestic market at $t$. In other words, it solves

$$\omega^{\sigma} + \phi W_{o,t+1}(\omega) = \Pi_{Dt}(\omega) + \phi W_{d,t+1}(\omega) \tag{30}$$

and in a neighborhood of this threshold both these options strictly dominate either export specialization or mixed exporting:

$$W_{ot} = \max\{\omega^{\sigma} + \phi W_{o,t+1}(\omega); W_{dt}(\omega)\} \tag{31}$$

at all dates $t$. (31) shows that the choice for these agents effectively reduces to a date $\tilde{t} \geq t$ when they enter the domestic market (and until $\tilde{t} - 1$ they remain in the traditional occupation); after $\tilde{t}$ the continuation value is the same. It follows that the optimal date of entry is the first $\tilde{t} \geq t$ at which $\omega^{\sigma} \leq \Pi_{D\tilde{t}}(\omega)$, which coincides with the choice made by myopic agents. Hence the threshold $\omega^*_{dt}$ is same as for a myopic agent.

The threshold $\omega^*_{et}$ for export specialization solves $W_{dt}(\omega) = W_{et}(\omega)$, i.e.,

$$\Pi_{Dt}(\omega) + \phi W_{d,t+1}(\omega) = \Pi_{Et}(\omega) + \phi W_{e,t+1}(\omega) \tag{32}$$

Since the corresponding continuation values $W_{d,t+1}(\omega), W_{e,t+1}(\omega)$ of specializing in the domestic and export markets will typically differ, this threshold will typically vary with $\phi$ even for small values of $\phi$. The threshold is of course continuous in $\phi$, so the expression for the export propensity in the text is an approximation for the true threshold for small values of $\phi$. Observe also that the greater the difference between growth of market or network size in the domestic and export markets between $t$ and $t+1$, the greater is the corresponding difference in change in the value of domestic specialization $\Pi_{D,t+1}(\omega) - \Pi_{Dt}(\omega)$ versus export specialization $\Pi_{E,t+1}(\omega) - \Pi_{Et}(\omega)$, and the higher will be $\omega^*_{et}$, resulting in a lower export propensity at $t$. This is a dynamic extension of the domestic network overhang effect amplifying the latter when agents are non-myopic.

**3. Composition of firms:**

Pure exporters have been observed in many developing countries and we now proceed to document their presence in China. We do this with data from the Economic Census, available in 2004 and 2008. These data provide revenues for all manufacturing firms and can be matched with the Customs database. Those firms whose revenues exceed their exports are designated as mixed exporters. Those firms whose revenues match their exports are classified as pure exporters. The economic census is the most reliable data-source that we have at our disposal. Nevertheless, there will be inaccuracies in reported revenues. We thus allow for up to 10% slippage between revenues and exports when classifying a firm as a pure exporter. Finally, those firms that do not appear in the customs data are assumed to be domestic producers.

TABLE B1

COMPOSITION OF FIRMS

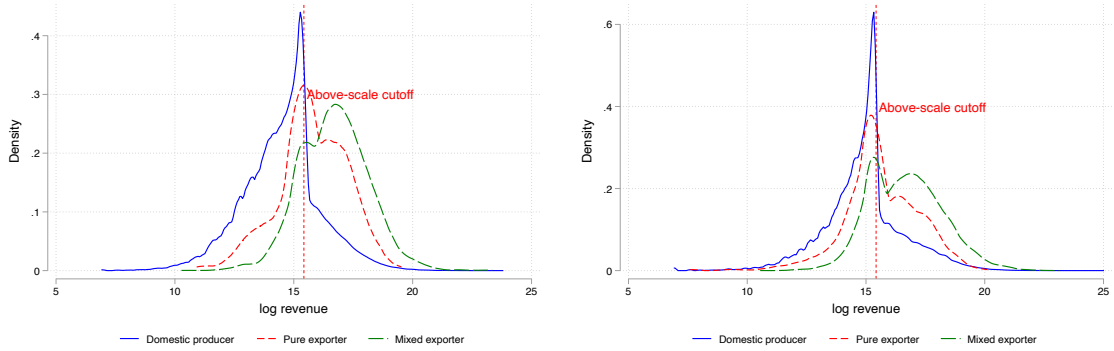
| Year | 2004 | | 2008 | |
|---|---|---|---|---|
| Type: | number | log revenue | number | log revenue |
| Domestic producer | 243,302 | 14.477 | 486,729 | 14.871 |
| Pure exporter | 615 | 15.763 | 2,180 | 15.592 |
| Mixed exporter | 4,396 | 16.544 | 10,933 | 16.440 |

Source: Economic Census (2004,2008) and Customs data.

Data restricted to manufacturing firms. Revenue measured in Yuan.

Table B1 describes the composition of firms in 2004 and 2008, based on the preceding classification. Export firms constitute a tiny fraction, around 2-3%, of all manufacturing firms and pure exporters comprise around 15% of all exporters. Notice that these firms can be ranked with respect to their revenue: domestic producers have the lowest revenues, followed by pure exporters and then mixed exporters. This ranking, which is also documented by Lu et al. (2014) using the Above Scale database, matches the ordering of firms in our model with respect to revenues (and ability). Figure B1 subjects the ranking to closer scrutiny by reporting the distribution of revenues for each type of firm. It can be seen that the distributions for domestic producers, pure exporters and mixed exporters, in that order, are increasingly shifted to the right.

The vertical line in Figure B1 marks the 5 million Yuan cutoff above which firms are selected into the Above Scale database, which is maintained by the National Bureau of Statistics and has been used in many previous studies. Above Scale firms are subjected to increased government oversight, which is presumably why there is bunching just below the threshold (especially for domestic firms). Firms in the Above Scale database are evidently highly selected, which is why we prefer the economic censuses and the SAIC databases




(a) 2004 (b) 2008

FIGURE B1.—Revenue Distribution
Source: Economic Census (2004,2008) and Customs Database.
Revenue measured in Yuan.

for our analyses. The SAIC inspection database, which we use for the analysis of network effects, also provides firm revenues. However, this is only for a sample of firms.

APPENDIX C: TESTING THE MODEL

## 1. Growth in network size with respect to duration

Based on Figure 1, we expect that the change in network size, in levels, from one period to the next will be increasing in network duration. This is what we observe in Columns 1-2 below. However, the sign of this association is reversed when network size is measured in logs, as seen in Columns 3-4. This is in line with what we observe in Table II.

TABLE C1

CHANGE IN NETWORK SIZE AND NETWORK DURATION

| Dependent variable: | change in network size | | change in log network size | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Network: | domestic | export | domestic | export |
| Duration | 23.503*** | 3.996*** | -0.005*** | -0.007*** |
| | (1.301) | (1.152) | (0.000) | (0.001) |
| Destination-time period effects effects | Yes | Yes | Yes | Yes |
| Observations | 1,317,332 | 60,125 | 1,317,332 | 60,125 |

Note: Network size is constructed from SAIC registration data and Customs data.
Network size is measured by the number of firms from the birth county in the destination prefecture.
The domestic network includes all firms, while the export network is restricted to exporting firms.
The change in network size (in levels) or in log network size is measured over successive periods.
Regressions are weighted by network size to be consistent with Table II, which is estimated at the firm level.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

**2. Constructing the shift-share instrument**

This instrument predicts the entry of firms from birth county $j$ into prefecture $k$ in time period $t$ based on agricultural productivity shocks in the birth county. It is constructed in the following steps:

**Step 1**: To construct the "shift" of the shift-share instrument, we calculate a crop-specific price shock for 11 crops that account for 96 percent of cultivated area in China. Agricultural Producer Prices (APP) at the "farm gate" are available for each producing country in USD between 1991 and 2016 from the FAO. Following Imbert et al. (2022), the world price of each crop $c$ is the average price across countries (excluding China) weighted by their yearly share of global exports. As in Imbert et al., the crop price shock, $\epsilon_{ct}$, is calculated by estimating the following equation:

$$\log P_{c,t} = \theta \log P_{c,t-1} + \eta_t + \nu_c + \epsilon_{ct}.$$

**Step 2**: To construct the first (inner) component of the "share" in the shift-share instrument, we construct a weight for each crop that reflects its contribution to total agricultural output, by value, in county $j$. The weighted sum of the crop price shocks then provides us with a measure of the income shock in county $j$ in year $t$:

$$S_{jt} = \sum_c \left( \frac{\overline{P}_c \cdot \overline{A}_{cj} \cdot y_{cj}}{\sum_c \overline{P}_c \cdot \overline{A}_{cj} \cdot y_{cj}} \right) \epsilon_{ct}$$

where $\overline{P}_c$ is the world price of crop $c$ in a reference year (1997), $\overline{A}_{cj}$ is the acreage allocated to crop $c$ in county $j$ in that year, and $y_{cj}$ is the potential crop yield (obtained from the FAO-GAEZ database). The acreage statistic is obtained from the 2000 World Census of Agriculture (WCA), which provides a geocoded map of harvest area for each crop at a 30 arc-second (approximately 10 km.) resolution. We aggregate the harvest areas to the county level to construct the acreage statistic. We choose 1997 as the reference year when constructing the crop weights because the WCA provides acreage in that year for China.

**Step 3**: The decision to establish a firm is a major decision that is unlikely to be determined by a single income shock. We thus assume that firm entry in year $t$ from county $j$ is determined by the average of the income shocks in that year and the preceding two years:

$$AS_{jt} = \frac{1}{3} \sum_{\tau=t-2}^{t} S_{j\tau}.$$

**Step 4**: The entering firms from birth county $j$ are then "distributed" across destination prefectures, $k$, by dividing the county-level average income shock by distance, $d_{jk}$, plus one. If a firm locates in its birth prefecture, the distance is set to zero. If not, the distance is measured from the centroid of the birth county to the centroid of the destination prefecture. The standard gravity equation in the New Economic Geography literature; e.g. Tombe and Zhu (2019) is specified as follows:

$$\log\left(\frac{n_{jkt}}{n_{jt}}\right) = \eta_{jt} + \eta_{kt} + \kappa \log(d_{jk}) + \varepsilon_{jkt}.$$

To construct the shift-share instrument, we set the "migration" elasticity, $\kappa$, to be equal to minus one, instead of estimating it from the preceding equation:

$$IV = \frac{AS_{jt}}{(d_{jk}+1)}.$$

The instrument that we construct can be compared and contrasted with the instrument used by Imbert et al. (2022) in their analysis of labor migration and firm productivity in China. We follow Imbert et al. in steps (i) and (ii), except that the income shocks are constructed in the birth county rather than the origin prefecture. Where we depart from their approach is in the steps that follow: we compute the average of the history of income shocks in (iii) and we divide by distance, instead of using the initial entry level, to allocate the predicted flow of firms across destination prefectures in (iv). Both our instruments have a shift-share structure, but the structure is interpreted differently. Imbert et al. think of the income shock as the shift, implicitly assuming that the crop shares are exogenous, while allowing the initial migration shares across destinations to be endogenous. We think of the crop price shocks as the shifts, with the crop shares and the distance multiplier together constituting the shares. We treat all components of our instrument as exogenous, with the discussion that follows assessing the validity of the exclusion restriction for each of them.

### 3. Validating the shift-share instrument

In this section, we assess whether each component of the shift-share instrument satisfies the exclusion restriction. Estimates with the benchmark specification, using the shift-share instrument and with the full sample of firms are reported in Table C2, Column 1.

**(a) Agricultural price shocks**: One way in which agricultural price shocks could directly impact firm revenue is if they affect the local economy more broadly and firms are located in the birth county itself. We allow for this possibility by restricting the sample to firms located outside their birth county in Table C2, Column 2. As can be seen, the network size coefficient continues to be positive and significant, although it is smaller in size than the benchmark coefficient estimate in Column 1.

A second way in which agricultural price shocks could affect a firm's performance is if it is operating in that sector. We address this concern by dropping firms that are engaged in activities associated with agriculture, such as food processing. The estimates reported in Table C2, Column 3 are very similar to what we obtain with the full sample.

Finally, a third way in which agricultural price shocks could directly affect business is through the wealth channel. We account for this in Table C2, Column 4 by including the uninteracted agricultural income shock in the birth county as a covariate in the estimating equation. The income shock has a positive and significant direct effect on firm revenue, whereas our first-stage estimates in Table II indicate that it has a negative effect on firm entry and, with it, network size. These effects work in opposite directions and, hence, by ignoring a potential wealth effect in the benchmark specification, we are (if anything) reporting conservative estimates of the network size effect.

**(b) Distance adjustment**: Distance is a fixed characteristic and, hence, its direct effect on firm revenue is subsumed in the firm fixed effect. However, Goldsmith-Pinkham et al. (2020) note that its interaction with time, and the interaction of the "share" with time more generally, must also be considered when examining the validity of the shift-share instrument. As observed in Table C2, Column 5 the results are robust to including distance interacted with time effects in the estimating equation.

**(c) Crop shares**: The crop shares, like the distance multiplier, are fixed characteristics and, hence, their direct effect on firm revenue is subsumed in the firm fixed effect. As with distance, however, we must consider the possibility that the interaction of the shares with time directly determines firm revenues when examining the validity of the shift-share instrument.

To address the preceding concerns, we take advantage of the fact that if the crop shares are exogenous, then the shift-share instrument that we construct is "equivalent" to using the shares associated with each crop, interacted with time effects, as independent instruments for network size (Goldsmith-Pinkham et al., 2020). If treatment effects are heterogeneous, then the instrumental variable estimates could vary by crop. However, Goldsmith-Pinkham

TABLE C2

ROBUSTNESS CHECK FOR SHIFT-SHARE IV

| Dependent variable: | log domestic revenue | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Sample of firms: | all | outside birth county | excluding agricultural processing | all | all |
| Log network size | 1.540*** | 0.634*** | 1.521*** | 1.602*** | 1.548*** |
| | (0.139) | (0.191) | (0.137) | (0.156) | (0.144) |
| Agriculture income shock | – | – | – | 0.149** | – |
| | | | | (0.067) | |
| Distance-time effects | No | No | No | No | Yes |
| Firm fixed effects | Yes | Yes | Yes | Yes | Yes |
| Destination-time period effects | Yes | Yes | Yes | Yes | Yes |
| Kleibergen-Paap F | 35.22 | 11.80 | 35.55 | 32.28 | 29.82 |
| Observations | 5,065,050 | 2,846,975 | 4,947,751 | 5,065,050 | 5,065,050 |

Note: network size is constructed from SAIC registration data.
Revenue is constructed from SAIC inspection data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variable is thus measured by the growth in (domestic) network size.
Instrument for the growth in network size: birth county agricultural shocks.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

TABLE C3

TESTING THE EXOGENEITY OF THE CROP SHARES: SHIFT-SHARE IV

| Dependent variable: | log domestic revenue | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Crop used in IV: | all | maize | potato | wheat | rapeseed | sorghum | soybean | rice |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Log network size | 1.373*** | 1.917*** | 2.157*** | 1.261*** | 1.214*** | 1.449*** | 1.383*** | 1.560*** |
| | (0.087) | (0.207) | (0.249) | (0.134) | (0.127) | (0.261) | (0.110) | (0.076) |
| Firm fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Destination-time effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Kleibergen-Paap F | 8.06 | 8.80 | 3.93 | 4.10 | 5.51 | 3.19 | 7.96 | 9.24 |
| Share | – | 0.33 | 0.12 | 0.10 | 0.12 | 0.01 | 0.12 | 0.07 |
| Weight | – | 0.41 | 0.24 | 0.13 | 0.11 | 0.05 | 0.04 | 0.03 |
| Observations | 5,065,050 | 5,065,050 | 5,065,050 | 5,065,050 | 5,065,050 | 5,065,050 | 5,065,050 | 5,065,050 |

Note: network size is constructed from SAIC registration data.
Revenue is constructed from SAIC inspection data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variable is thus measured by the growth in (domestic) network size.
Instrument for the growth in network size: crop share interacted with time effects.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

et al. note that substantial heterogeneity or a single outlying crop is still a concern with regard to the validity of the shift-share instrument. Table C3 reports results with domestic revenue as the dependent variable, using the share for each crop interacted with time effects

(and the distance adjustment) as instruments for network size. We report estimates with all 7 of the 11 crops that have a positive Rotemberg weight, a statistic derived by Goldsmith-Pinkham, Sorkin and Swift that measures the contribution of a given crop to the shift-share instrument. Among these crops, maize, potato, repeseed, and rice have the largest weights, together accounting for 79.2 percent of the variation in the instrument and 63.1 percent of the harvesting acreage. The network effects estimated separately with each of these crops are positive, significant, and similar in magnitude to each other and to the benchmark estimates with the shift-share instrument in Table C2, Column 1.

## 4. Validating the duation instrument

Entrepreneurial experience will be mechanically correlated with network duration. If experience determines firm revenue directly, then this will potentially undermine the domestic and export network duration instruments in the domestic and export revenue equations, respectively. We address this possibility below by including entrepreneurial experience in the first-differenced revenue equation. Although the estimated network effects do decline in size, they continue to be positive and highly statistically significant.

TABLE C4

TESTING THE EXOGENEITY OF THE DURATION IV: INDIVIDUAL EXPERIENCE EFFECT

| | (1) log domestic revenue | (2) log export revenue |
|---|---|---|
| Log network size | 0.366*** | 0.598*** |
| | (0.055) | (0.136) |
| Entrepreneurial experience | -0.042*** | -0.031*** |
| | (0.001) | (0.001) |
| Firm fixed effects | Yes | Yes |
| Destination-time period effects | Yes | Yes |
| Kleibergen-Paap F | 92.05 | 26.01 |
| Observations | 5,065,050 | 281,091 |

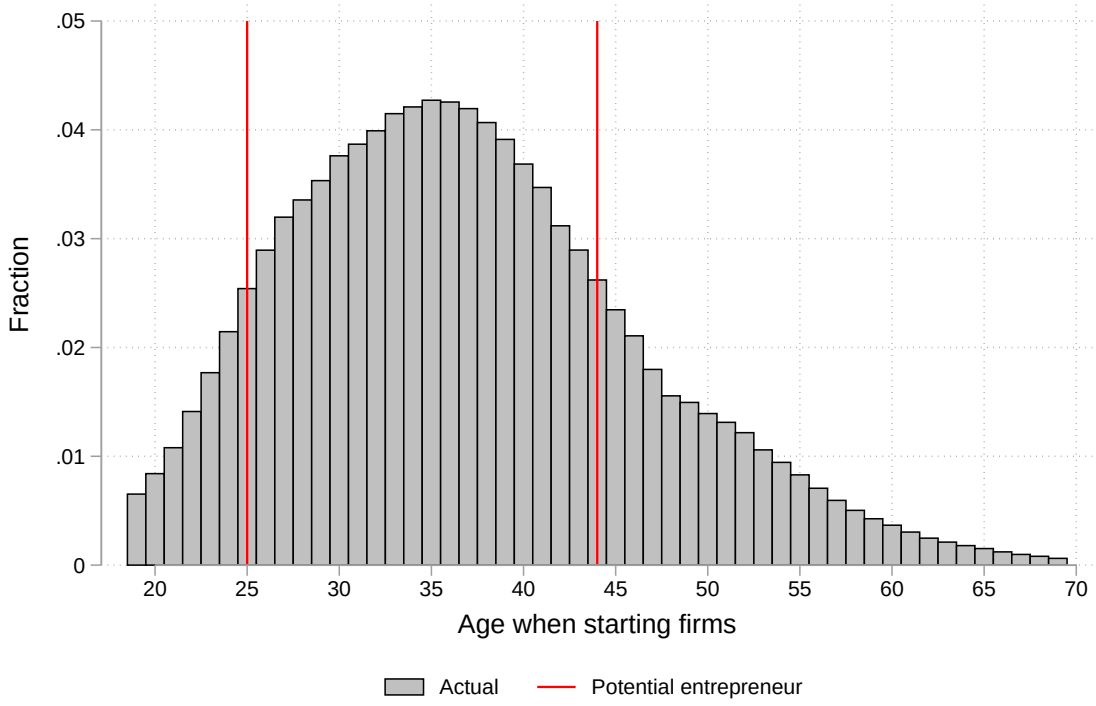
Note: network size is constructed from SAIC registration data and Customs data.
Revenue is constructed from SAIC inspection data and Customs data.
Firm fixed effects are purged by first-differencing prior to estimation.
The modified network variable is thus measured by the growth in its size: $\log n_{jk,t-1} - \log n_{jk,t-2}$ for the domestic network and $\log n_{ejk,t-1} - \log n_{ejk,t-2}$ for the export network.
Instruments for the growth in domestic network size: birth county agricultural shocks, domestic network duration.
Instruments for the growth in export network size: export network duration, domestic network duration.
Entrepreneurial experience is measured by the difference between the current financial year and the year of establishment of the firm.
Standard errors clustered at the birth county level in parentheses. ** significant at 5%, *** at 1%.

APPENDIX D: QUANTITATIVE ANALYSIS

## 1. Computing the number of potential entrepreneurs

The SAIC registration database provides the gender and age for a subset of legal representatives. Among those individuals who report their gender, 79 percent are men. In addition, we see in Appendix Figure D1 that most (male) entrepreneurs in the registration database were aged 25-44 when their firm was established. The number of potential entrepreneurs from a given county at a particular point in time is thus specified to be the number of men currently aged 25-44 who were born in that county. This statistic is derived in each year using a one percent sample from the 1990 census. Large-scale internal migration only commenced in China in the 1990's and, hence, the gender and age distribution of county residents in the 1990 census can be used, without modification.

FIGURE D1.—Age Distribution of Male Entrepreneurs




Source: SAIC registration database.

*Co-editor [Name Surname; will be inserted later] handled this manuscript.*